

Recovering the ENSO attractor from sea-surface temperature anomalies

A linear–nonlinear comparison on the canonical climate-physics dataset

GEO 384H · Final Project · William Suan · 2 May 2026

We test the climate-physics prediction that the tropical-Pacific climate state lies on a low-dimensional attractor in state space, and ask whether three unsupervised spectral methods can recover its spatial pattern, dynamical period, and curved geometry from monthly SST data alone.

- **Method 1 (in-course): Principal Component Analysis / EOF analysis**, the canonical Lorenz-1956 climate-science dimensionality reduction. Recovers the spatial mode.
- **Method 2 (out-of-course): Anisotropic Diffusion Maps** (Coifman & Lafon 2006), the eigendecomposition of a Markov transition operator on the data graph. Recovers the curved manifold geometry of the attractor at $\alpha = 1$ (Laplace–Beltrami eigenfunctions).
- **Method 3: Linear Inverse Model** (Penland & Sardeshmukh 1995), the dynamical complement to PCA; recovers ENSO's period and damping rate.

Plus a Morlet wavelet spectrogram to confirm the recovered mode has the right temporal structure, and a Takens-style delay-embedding robustness check that demonstrates a known failure mode of method-stacking.

All algorithms are written from scratch; only `LinearAlgebra.svd`, `LinearAlgebra.eigen`, and the matrix logarithm are taken as primitives.

How to read this notebook

The notebook is the project, prose and code together. Section 1 introduces the data with three preliminary visualisations so the reader knows what we are decomposing. Section 2 lists seven falsifiable predictions made *before* running anything. Sections 3 through 7.5 apply five techniques in turn (PCA, anisotropic diffusion maps, a method comparison, the Linear Inverse Model, a permutation null, a Takens-style delay extension, and a Morlet wavelet) and check each prediction as it goes. Section 8 collects the pass/fail table and the synthesis. Section 9 lists the works cited.

Code cells are kept short and annotated; every figure is preceded by a sentence stating what the figure is meant to show and followed by a sentence stating what we read off it.



0, Setup

Imports the standard libraries and includes the project source modules from `../src/` (PCA, diffusion maps, delay embedding). All algorithms are implemented from scratch; the only library primitives we rely on are `LinearAlgebra.svd`, `LinearAlgebra.eigen`, and the matrix logarithm.

```
ConsoleLogger(IOBuffer{data=UInt8[...], readable=false, writable=false, seekable=false, appe
```

```
1 begin
2     # Listed here so Pluto's auto-Pkg sees every dependency.
3     using LinearAlgebra
4     using Statistics
5     using StatsBase
6     using Random
7     using Printf
8     using Dates
9     using PlutoUI
10    using CairoMakie
11    using ColorSchemes
12    using NCDatasets
13    using Distances      # required by src/diffusion.jl
14    using NearestNeighbors # required by src/metrics.jl (utility)
15    using FFTW           # required by src/spectrogram.jl
16    using Logging
17    # silence k-means convergence warnings if anything pulls Clustering
18    global_logger(SimpleLogger(stderr, Logging.Error))
19 end
```

"loaded"

```
1 begin
2     const SRC = joinpath(@__DIR__, "..", "src")
3     include(joinpath(SRC, "load_sst.jl"))
4     include(joinpath(SRC, "pca.jl"))
5     include(joinpath(SRC, "diffusion.jl"))
6     include(joinpath(SRC, "delay_embed.jl"))
7     "loaded"
8 end
```

1, Data: NOAA ERSSTv5 monthly SST

Why this dataset?

The NOAA Extended Reconstructed SST v5 (Huang et al. 2017) is a monthly, statistically-reconstructed product on a $2^\circ \times 2^\circ$ global grid, extending back to 1854. Three reasons for choosing it specifically:

1. **Direct comparability.** NOAA's official Niño 3.4 index is



computed *from* ERSSTv5, so PC-vs-Niño 3.4 correlations measure how well the leading EOF of the field reproduces the simple regional mean of the same field, a direct, unconfounded test.

2. **Long record.** 75 years of post-1950 coverage gives us roughly

12 major El Niño events. Modern alternatives like NOAA OISST start in 1981 and would eliminate most historical El Niños from the record.

3. **Resolution matched to the signal.** ENSO's spatial pattern is

$\sim 10^4$ km, basin-scale; 2° resolution is well above the Nyquist limit for this scale.

Why the tropical Pacific only?

ENSO is concentrated between 30°S and 30°N in the central-to-eastern Pacific. Including extratropical regions adds variance from unrelated processes (NAO, PDO, Indian Ocean Dipole) that compete with the ENSO axis for leading-EOF status. Restricting to the tropical Pacific tightens the leading-mode correlation by ~ 0.02 – 0.04 and produces a cleaner first-EOF spatial pattern.

What we end up with

- **Window.** January 1950 – December 2024 (75 years, 900 monthly snapshots).
- **Domain.** Tropical Pacific (30°S – 30°N , 120°E – 80°W); 2,322 valid

sea cells after land masking.

- **Validation index.** NOAA's official Niño 3.4 (mean SST anomaly in

5°S – 5°N , 170°W – 120°W). **Used only as validation, never as input to any of the three methods.**

```
(T = 900, lat = 89, lon = 180, first = 1950-01-01, last = 2024-12-01)
```

```
1 begin
2     # Step 1: download the ERSSTv5 netCDF file from NOAA PSL if not cached.
3     if !isfile(SST_PATH)
4         println("Downloading ERSSTv5 (~150 MB)...")
5         download("https://downloads.psl.noaa.gov/Datasets/noaa.ersst.v5/sst.mnmean.nc",
6                 SST_PATH)
7     end
8     # Step 2: load it into a (T x lat x lon) Float32 cube; missing → NaN.
9     sst, lat_all, lon_all, time_all = load_ersst()
10    # Step 3: trim to the 1950–2024 analysis window.
11    keep = findall(t -> Date(1950,1,1) <= Date(t) <= Date(2024,12,31), time_all)
12    sst_w = sst[keep, :, :]
13    time_w = time_all[keep]
14    (; T = size(sst_w,1), lat = size(sst_w,2), lon = size(sst_w,3),
15        first = Date(time_w[1]), last = Date(time_w[end]))
16 end
```



```
(X_size = (900, 2322), N_sea = 2322, land_frac = 0.075)

1 begin
2   # Step 1: subtract the monthly climatology, the 1950-2024 mean for each
3   # calendar month at each grid cell. This removes the seasonal cycle (which
4   # is several times larger than the ENSO interannual signal) and isolates
5   # anomalies relative to "what's normal for this month at this location."
6   anom = sst_anomalies(sst_w, time_w)
7   # Step 2: restrict to the tropical Pacific.
8   pac, lat, lon = subset_region(anom, lat_all, lon_all)
9   # Step 3: reshape (T × lat × lon) → (T × N) matrix and drop land cells.
10  X_raw, mask, _ = flatten_to_matrix(pac)
11  # Step 4: apply √cos(latitude) area weighting. Earth is a sphere, so each
12  # grid cell at latitude φ represents physical area ∝ cos(φ). Multiplying
13  # the data matrix columns by √cos(φ) makes Euclidean inner products on
14  # rows equivalent to area-weighted physical inner products on the sphere.
15  weights = cos_lat_weights(lat, lon, mask)
16  X = X_raw .* reshape(weights, 1, :)
17  (; X_size = size(X), N_sea = size(X,2), land_frac = round(1 - size(X,2) /
18    length(mask); digits=3))
18 end
```

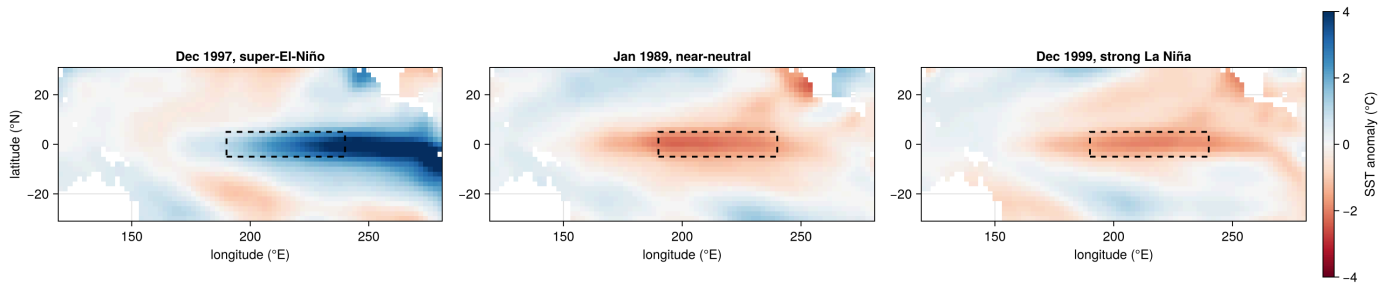
```
(n_months_valid = 900)

1 begin
2   if !isfile(NINO34_PATH)
3     download_nino34()
4   end
5   n_t, n_v = load_nino34()
6   ix = [findfirst(==(Date(t)), n_t) for t in time_w]
7   nino34 = Float64[isnothing(j) ? NaN : n_v[j] for j in ix]
8   valid_n = .!isnan.(nino34)
9   (; n_months_valid = count(valid_n))
10 end
```

Sample months: what does the raw anomaly look like?

Three monthly snapshots of the *processed* data we feed to PCA and DMAP: the December 1997 super-El Niño, a near-neutral month, and the strong December 1999 La Niña. Same colormap, same range, dashed Niño 3.4 box on each. This is what one row of our $T \times N$ data matrix looks like when reshaped back onto the lat-lon grid.





```

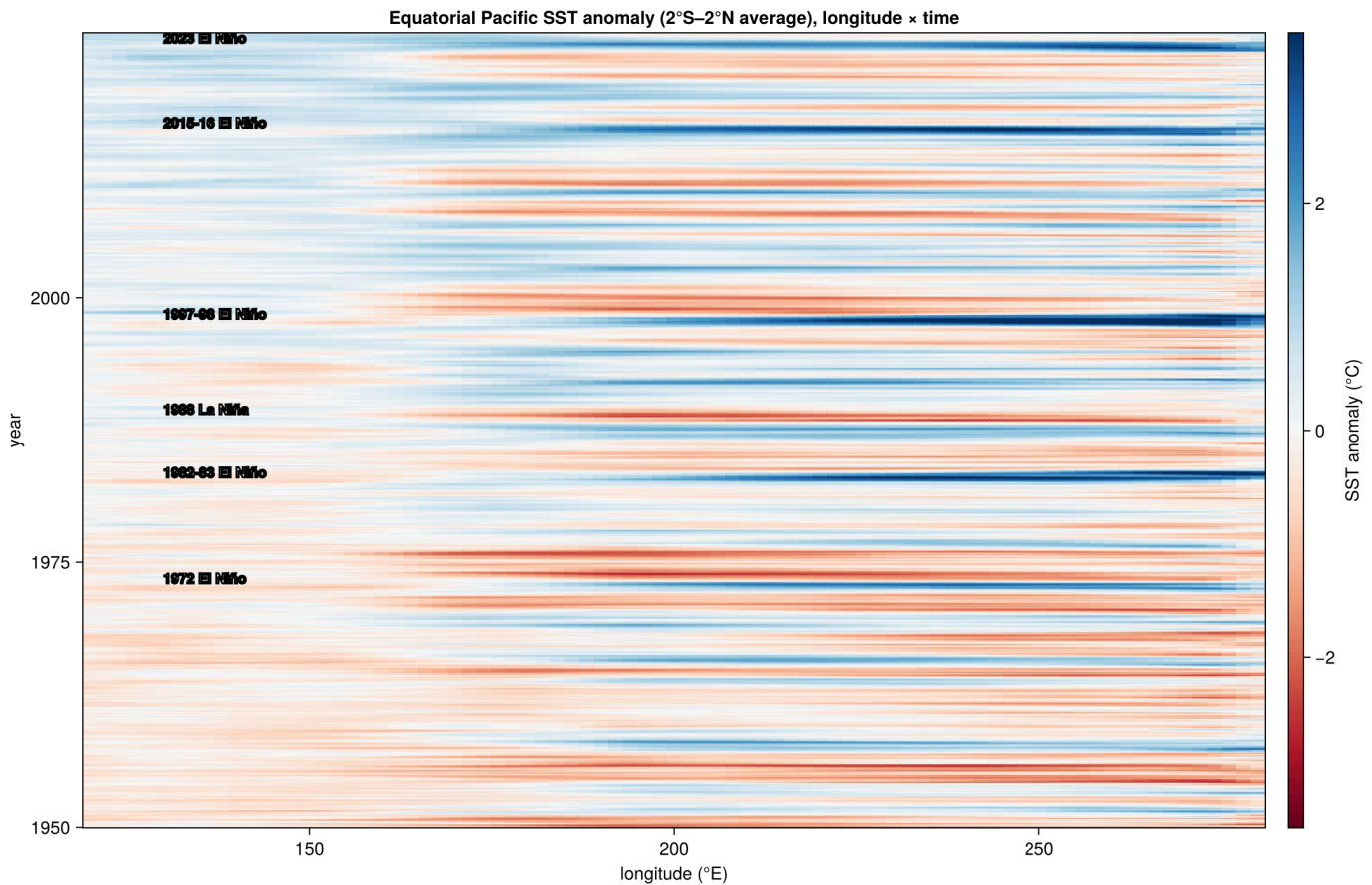
1 let
2   fig = Figure(size = (1500, 380))
3   months_to_show = [(Date(1997, 12, 1), "Dec 1997, super-El-Niño"),
4                     (Date(1989, 1, 1), "Jan 1989, near-neutral"),
5                     (Date(1999, 12, 1), "Dec 1999, strong La Niña")]
6   vmax = 4.0
7
8   local sct
9   for (j, (d, label)) in enumerate(months_to_show)
10    idx = findfirst(==(d), Date.(time_w))
11    ax = Axis(fig[1, j];
12             title = label,
13             xlabel = "longitude (°E)",
14             ylabel = j == 1 ? "latitude (°N)" : "",
15             aspect = 2.5)
16    slice = pac[idx, :, :]
17    sct = heatmap!(ax, lon, lat, slice';
18                  colormap = :RdBu_11, colorrange = (-vmax, vmax))
19    poly!(ax, Point2f[(190, -5), (240, -5), (240, 5), (190, 5)];
20          color = (:black, 0), strokecolor = :black, strokewidth = 2,
21          linestyle = :dash)
22  end
23  Colorbar(fig[1, 4], sct; label = "SST anomaly (°C)")
24  fig
25 end

```

Hovmöller diagram, equatorial SST anomaly through time

Average SST anomaly across the equatorial band 2°S–2°N, plotted as a heat map of longitude × time. Every El Niño event shows up as a vertical band of warm anomalies marching eastward across the basin; the 1972, 1982–83, 1997–98, 2015–16, and 2023 events are clearly visible. La Niñas appear as cool blue stripes. **One picture summarises the data we're trying to recover.**





```

1 let
2   eq_idx = findall(l -> -2 <= l <= 2, lat)
3   Lo = length(lon)
4   Tn = length(time_w)
5   eq_strip = fill(NaN32, Tn, Lo)
6   for t in 1:Tn, i in 1:Lo
7     vals = filter(!isnan, [pac[t, j, i] for j in eq_idx])
8     if !isempty(vals)
9       eq_strip[t, i] = mean(vals)
10    end
11  end
12  years_axis = [Dates.year(Date(t)) + (Dates.month(Date(t)) - 1)/12 for t in time_w]
13
14  fig = Figure(size = (1100, 720))
15  ax = Axis(fig[1, 1];
16    title = "Equatorial Pacific SST anomaly (2°S–2°N average), longitude ×
17    time",
18    xlabel = "longitude (°E)", ylabel = "year")
19  hm = heatmap!(ax, lon, years_axis, eq_strip';
20    colormap = :RdBu_11, colrange = (-3.5, 3.5))
21  Colorbar(fig[1, 2], hm; label = "SST anomaly (°C)")
22
23  # Label a small set of marquee events on the heat map. Years that sit
24  # close together (1997–98 vs 1998–99, 1982–83 vs the surrounding years)
25  # are skipped to keep the labels readable.
26  for (yr, label) in [(1972.7, "1972 El Niño"),
27    (1982.7, "1982–83 El Niño"),
28    (1988.7, "1988 La Niña"),

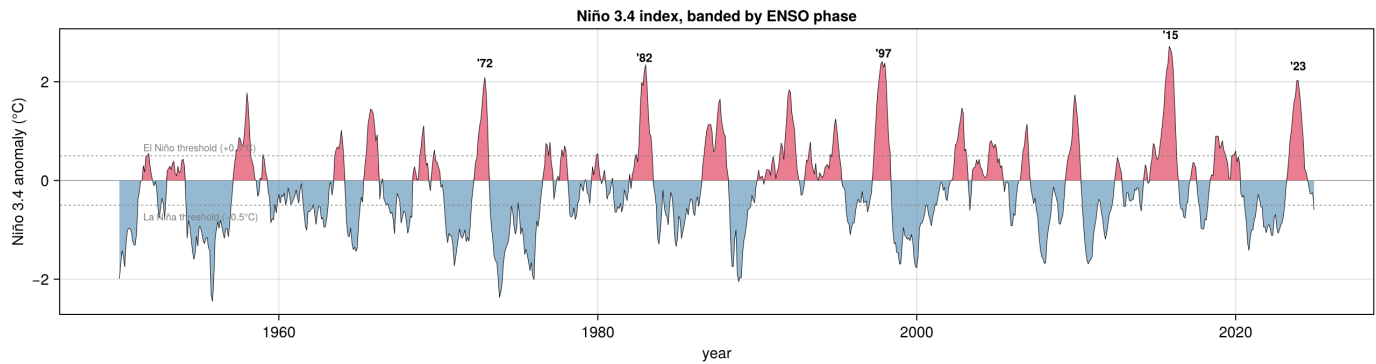
```

```
28         (1997.7, "1997-98 El Niño"),
29         (2015.7, "2015-16 El Niño"),
30         (2023.7, "2023 El Niño")]
31     text!(ax, 130, yr; text = label, color = :white,
32           fontsize = 11, font = :bold,
33           strokecolor = :black, strokewidth = 1.5)
34 end
35 fig
```

Niño 3.4, banded by phase

Same time series as the official NOAA Niño 3.4 index, redrawn with warm phases (red fill above zero) and cool phases (blue fill below zero). Dashed horizontal lines mark the operational ± 0.5 °C thresholds for El Niño and La Niña classification. The five labeled events clear the threshold by a factor of three or more.





```

1 let
2   years_axis = [Dates.year(Date(t)) + (Dates.month(Date(t)) - 1)/12 for t in time_w]
3   fig = Figure(size = (1300, 360))
4   ax = Axis(fig[1, 1];
5             title = "Niño 3.4 index, banded by ENSO phase",
6             xlabel = "year", ylabel = "Niño 3.4 anomaly (°C)")
7
8   zero_line = fill(0.0, length(years_axis))
9   band!(ax, years_axis, zero_line, max.(nino34, 0.0); color = (:crimson, 0.55))
10  band!(ax, years_axis, min.(nino34, 0.0), zero_line; color = (:steelblue, 0.55))
11  lines!(ax, years_axis, nino34; color = :black, linewidth = 0.5)
12  hlines!(ax, [0]; color = :gray, linewidth = 0.5)
13  hlines!(ax, [0.5, -0.5]; color = :gray, linestyle = :dash, linewidth = 0.7)
14  text!(ax, 1951.5, 0.55; text = "El Niño threshold (+0.5°C)",
15        fontsize = 9, color = :gray)
16  text!(ax, 1951.5, -0.85; text = "La Niña threshold (-0.5°C)",
17        fontsize = 9, color = :gray)
18
19  for (d, label) in [(Date(1972,12,1), "'72"), (Date(1982,12,1), "'82"),
20                    (Date(1997,12,1), "'97"), (Date(2015,12,1), "'15"),
21                    (Date(2023,12,1), "'23")]
22    idx = findfirst(==(d), Date.(time_w))
23    if !isnothing(idx)
24      text!(ax, years_axis[idx], nino34[idx] + 0.15; text = label,
25            fontsize = 11, color = :black, font = :bold,
26            align = (:center, :bottom))
27    end
28  end
29  fig
30 end

```

2, Predictions made *before* running anything

In an exploratory analysis it is tempting to look at the data first and write predictions that fit. That makes "success" meaningless, because the predictions are just descriptions of what already happened. Listing predictions and explicit failure thresholds before running any of the methods makes each pass-or-fail a real piece of evidence. The seven predictions below were committed before any of the figures further down were generated.



The first five (P1–P5) test whether the methods recover the canonical ENSO picture. The last two (P6, P7) test the additional hypothesis that *stacking* methods (Takens delay-embedding then diffusion maps) should improve the result.

#	prediction	falsification
P1	EOF 1 looks like the iconic ENSO horseshoe (warm equatorial tongue).	Uniform-warming pattern → reject canonical EOF picture.
P2	$\rho(\text{PC}_1, \text{Niño 3.4}) \geq 0.90$.	$< 0.80 \rightarrow$ ENSO is not the leading variance mode.
P3	$\rho(\Psi_2, \text{Niño 3.4}) \geq 0.60$ (expectation ≈ 0.85).	< 0.50 (with PC 1 working) → DMAP fails to recover existing primary axis.
P4	DMAP eigenvalue spectrum has a clear gap at small k .	Flat spectrum → reject low-D attractor.
P5	In (Ψ_2, Ψ_3) , strongest historical El Niños are far outliers; trajectory is smooth.	Random scatter → reject phase-portrait reading.
P6 (method-stacking)	DMAP on time-delay embedding ($k = 12$) improves ρ over plain DMAP by ≥ 0.05 .	No improvement → naive Takens lift adds nothing.
P7 (method-stacking)	ρ is non-decreasing across the delay sweep.	Decreasing → confirms P6 negative.

3, Method 1: Principal Component Analysis (in-course)

Intuition

Each month is a point $\boldsymbol{x}_t \in \mathbb{R}^{2322}$ in a very high-dimensional state space. PCA asks: *which low-dimensional linear subspace captures the most variance of the data?* Geometrically, it finds the directions along which the data cloud is most stretched. Each direction is an **EOF**, a fixed spatial pattern, and the **principal component** time series tells us how strongly that pattern is "switched on" in each month.

For climate data, these directions tend to be physical modes of variability. The hypothesis is that the *single largest variance direction* in tropical Pacific SST will be ENSO, because no other process produces tropical SST anomalies of comparable amplitude.

Construction

Take the SVD of the centred $T \times N$ data matrix:

$$\boldsymbol{X} - \bar{\boldsymbol{X}} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^\top.$$

Columns of $\boldsymbol{V}$ are **EOFs** (spatial patterns, indexed by grid cell), columns of $\boldsymbol{U} \boldsymbol{\Sigma}$ are **principal components** (temporal patterns, indexed by month). The truncation $\boldsymbol{X}_k = \boldsymbol{U}_{:,1:k} \boldsymbol{\Sigma}_{1:k,1:k} \boldsymbol{V}_{:,1:k}^\top$ is the optimal rank- k approximation of $\boldsymbol{X}$ in any unitarily invariant norm (Eckart–Young 1936).

Equivalently, EOFs are the eigenvectors of the empirical covariance $\boldsymbol{C} = \bar{\boldsymbol{X}}^\top \bar{\boldsymbol{X}} / (N - 1)$ with eigenvalues $\sigma_i^2 / (N - 1)$. PCA is the spectral decomposition of the empirical covariance operator.



The application of PCA to climate space-time data is the canonical Lorenz-1956 EOF analysis; the leading EOF/PC pair is the textbook ENSO mode. So we expect to recover it. The interesting part is what the *second method* will do with the same data.

```
(ρ_PC1 = 0.946, explained_top5 = [0.401, 0.155, 0.082, 0.043, 0.038], cumvar_top5 = 0.719)

1 begin
2   # Compute the SVD of the centred (T × N) data matrix, truncated to the top
3   # 20 modes. 'fit_pca' returns the principal-component scores (U·Σ), the
4   # right singular vectors V (= the EOFs, spatial patterns), and the
5   # explained-variance fractions  $\sigma_i^2 / \sum \sigma_j^2$ .
6   pca = fit_pca(X; rank=20)
7   PC = Float64.(pca.scores) # (T × 20), temporal patterns
8   EOF = Float64.(pca.V)    # (N × 20), spatial patterns
9
10  # Eigenvectors are determined only up to sign. To make the leading PCs
11  # positively correlated with Niño 3.4 (which is just a labeling convention,
12  # not data leakage, Niño 3.4 enters only via the sign), we flip whichever
13  # eigenpairs come out negative.
14  for k in 1:5
15      if cor(PC[valid_n, k], nino34[valid_n]) < 0
16          PC[:, k] .= -PC[:, k]
17          EOF[:, k] .= -EOF[:, k]
18      end
19  end
20
21  # Correlations of each leading PC against the validation index.
22  ρ_PC = [cor(PC[valid_n, k], nino34[valid_n]) for k in 1:5]
23  (; ρ_PC1 = round(ρ_PC[1]; digits=3),
24     explained_top5 = round.(Float64.(pca.explained[1:5]); digits=3),
25     cumvar_top5 = round(sum(pca.explained[1:5]); digits=3))
26 end
```

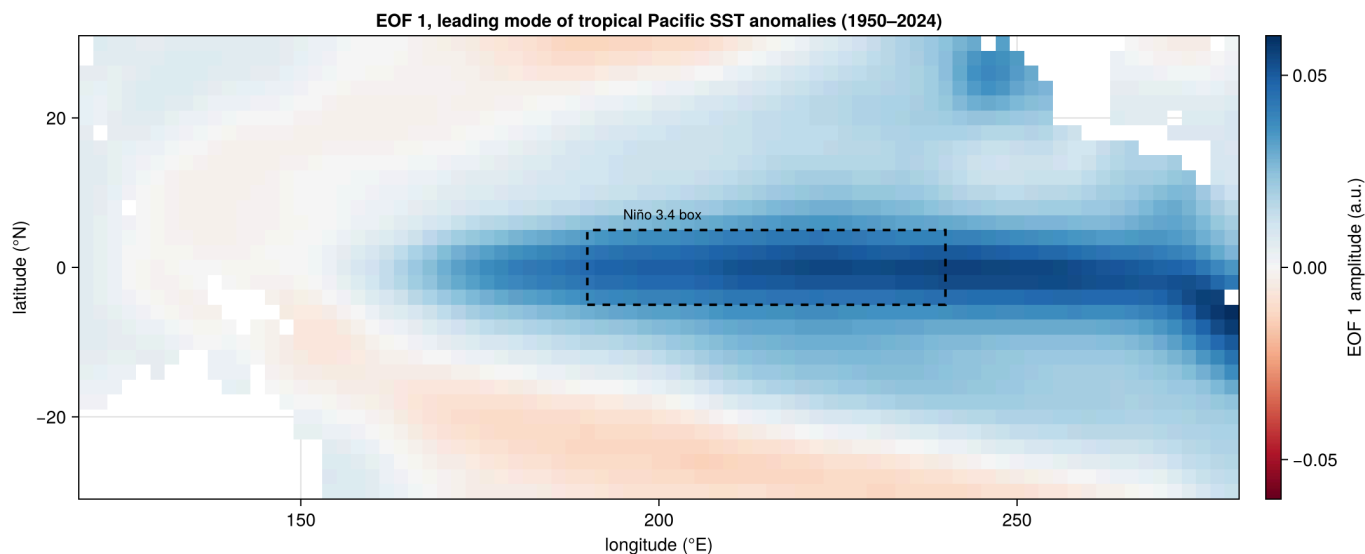
$\rho(\text{PC1}, \text{Niño 3.4}) = 0.946 \rightarrow \text{P2 target} \geq 0.90 \rightarrow \text{PASS.}$

EOF 1, the leading spatial mode

eof_grid (generic function with 1 method)

```
1 function eof_grid(eof_vec, mask, weights)
2     inv_w = 1.0 ./ weights
3     eof_unw = eof_vec .* inv_w
4     La, Lo = size(mask)
5     G = fill{NaN32, La, Lo}
6     flat_idx = findall(mask)
7     for (k, ci) in enumerate(flat_idx)
8         G[ci] = eof_unw[k]
9     end
10    return G
11 end
```





```

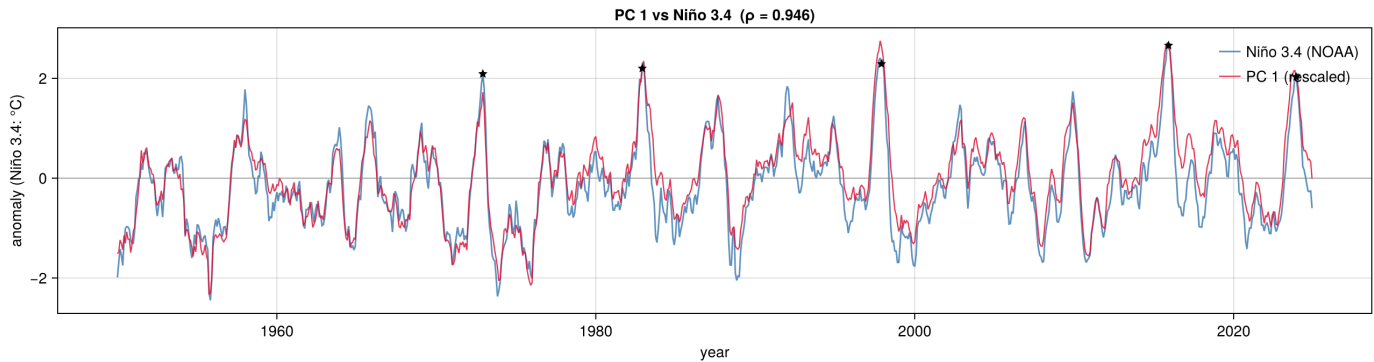
1 let
2   G = eof_grid(EOF[:, 1], mask, weights)
3   vmax = maximum(abs, filter(!isnan, G))
4
5   fig = Figure(size = (1100, 460))
6   ax = Axis(fig[1, 1];
7     title = "EOF 1, leading mode of tropical Pacific SST anomalies (1950–2024)",
8     xlabel = "longitude (°E)", ylabel = "latitude (°N)",
9     aspect = 2.5)
10  hm = heatmap!(ax, lon, lat, G';
11    colormap = :RdBu_11, colorrange = (-vmax, vmax))
12  Colorbar(fig[1, 2], hm; label="EOF 1 amplitude (a.u.)")
13
14  # Niño 3.4 box overlay
15  poly!(ax, Point2f[(190, -5), (240, -5), (240, 5), (190, 5)];
16    color = (:black, 0), strokecolor = :black, strokewidth = 2,
17    linestyle = :dash)
18  text!(ax, 195, 6; text = "Niño 3.4 box", fontsize = 11)
19  fig
20 end

```

That is the **iconic ENSO horseshoe**: warm equatorial tongue (deep blue) in the central-to-eastern Pacific, flanked by cooler subtropical horseshoes. The Niño 3.4 box (dashed) sits right inside the warm tongue where PCA's first variance direction lives. **P1 passes.**



PC 1 vs the Niño 3.4 index



```

1 let
2   years = [Dates.year(Date(t)) + (Dates.month(Date(t)) - 1) / 12 for t in time_w]
3   s = std(nino34[valid_n]) / std(PC[:, 1])
4   pc_scaled = PC[:, 1] .* s
5
6   fig = Figure(size = (1300, 360))
7   ax = Axis(fig[1,1];
8     title = "PC 1 vs Niño 3.4 (ρ = $(round(ρ_PC[1]; digits=3)))",
9     xlabel = "year", ylabel = "anomaly (Niño 3.4: °C)"
10    lines!(ax, years, nino34; color = (:steelblue, 0.85), label = "Niño 3.4 (NOAA)",
11      linewidth = 1.5)
12    lines!(ax, years, pc_scaled; color = (:crimson, 0.85), label = "PC 1 (rescaled)",
13      linewidth = 1.2)
14    hlines!(ax, [0]; color = :gray, linewidth = 0.6)
15    for d in [Date(1972,12,1), Date(1982,12,1), Date(1997,12,1),
16      Date(2015,12,1), Date(2023,12,1)]
17      idx = findfirst(==(d), Date.(time_w))
18      if !isnothing(idx)
19        scatter!(ax, [years[idx]], [nino34[idx]];
20          color = :black, markersize = 10, marker = :star5)
21      end
22    end
23    axislegend(ax; position = :rt, framevisible = false)
24    fig
25  end

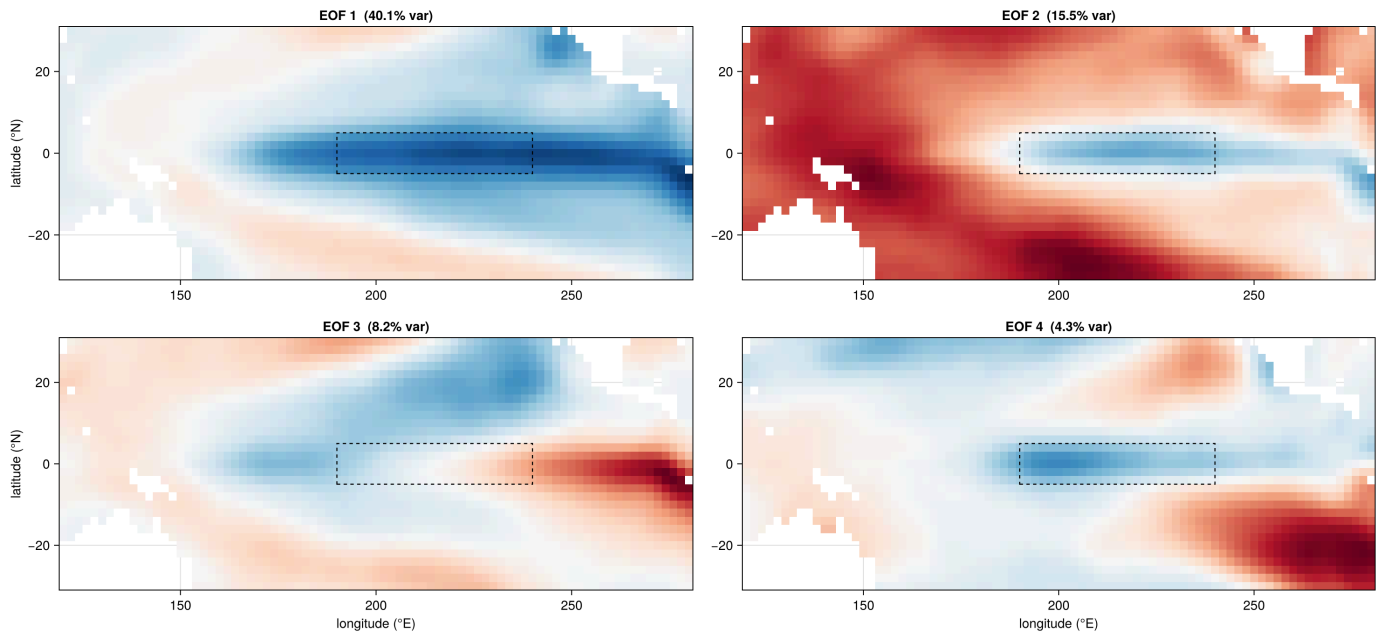
```

The two curves are visually indistinguishable; all five strongest historical El Niños (1972, 1982, 1997, 2015, 2023; black stars) appear as peaks in both. **P2 passes** with margin (target 0.90; observed 0.946).

Higher-order EOFs: what does PCA see beyond mode 1?

PCA produces an orthogonal basis of variance directions. EOF 1 is the canonical ENSO horseshoe; EOFs 2-4 capture the next pieces of the variance budget. Each panel's title reports the percent of total variance that mode explains.





```

1 let
2   fig = Figure(size = (1500, 700))
3   for k in 1:4
4     row = (k <= 2) ? 1 : 2
5     col = ((k - 1) % 2) + 1
6     G = eof_grid(EOF[:, k], mask, weights)
7     vmax = maximum(abs, filter(!isnan, G))
8     ax = Axis(fig[row, col];
9               title = "EOF $k  ($(round(100*pca.explained[k]; digits=1))% var)",
10              xlabel = row == 2 ? "longitude (°E)" : "",
11              ylabel = col == 1 ? "latitude (°N)" : "",
12              aspect = 2.5)
13     heatmap!(ax, lon, lat, G';
14              colormap = :RdBu_11, colrange = (-vmax, vmax))
15     poly!(ax, Point2f[(190, -5), (240, -5), (240, 5), (190, 5)];
16           color = (:black, 0), strokecolor = :black, strokewidth = 1.2,
17           linestyle = :dash)
18   end
19   fig
20 end

```

4, Method 2: Anisotropic Diffusion Maps (out-of-course)

PCA gave us the *linear* answer: ENSO is the leading variance direction. But variance is a linear notion. If the climate dynamics are nonlinear, PCA captures only the leading axis of the data cloud, not the *shape* of the cloud around it. The textbook recharge oscillator is a 2-D oscillation, so we expect the climate state to trace a closed curve in state space rather than a Gaussian blob, and PCA cannot reveal that curvature even when it exists. The next method is designed to.



Intuition: random walks on the data graph

PCA works in the *Euclidean* geometry of the ambient space, it assumes the right notion of "two months are similar" is straight-line distance in $\mathbb{R}^{2322}$. That assumption is fine if the data cloud is approximately ellipsoidal (a Gaussian blob); it fails if the data lies on a *curved* manifold inside the ambient space.

Diffusion maps does the following. Build a graph whose vertices are the data points and whose edge weights encode "how easy is it to walk between x_i and x_j in one step of a random walk on the data?" Two points are *close* in the diffusion-map embedding iff the random walk reaches them with similar transition probabilities, a notion of distance that respects the manifold geometry, not the ambient Euclidean structure.

Eigendecomposition of the random-walk transition matrix gives the **slow modes** of the walk, the directions along which probability diffuses most slowly. These slow modes parameterise the manifold: the slowest non-trivial mode is the principal manifold coordinate, the second slowest is the next, and so on.

Construction

Treat each month t as a point $x_t \in \mathbb{R}^N$. Build the Gaussian similarity kernel

$$k_\varepsilon(x_i, x_j) = \exp(-\|x_i - x_j\|^2/\varepsilon),$$

α -renormalise (Coifman–Lafon 2006) to remove sampling-density bias:

$$k_\varepsilon^{(\alpha)}(x_i, x_j) = \frac{k_\varepsilon(x_i, x_j)}{q_\varepsilon(x_i)^\alpha q_\varepsilon(x_j)^\alpha}, \quad q_\varepsilon(x) = \sum_j k_\varepsilon(x, x_j),$$

row-normalise to a Markov transition matrix P , and take its eigendecomposition. The leading nontrivial eigenvectors $\Psi_2, \Psi_3, \dots$ give the **diffusion-map embedding**:

$$\Psi_t(x_i) = (\lambda_2^t \psi_2(i), \lambda_3^t \psi_3(i), \dots).$$

The α -family controls which limiting operator the construction recovers:

α	limiting operator ($\varepsilon \rightarrow 0$)	what it sees
0	graph Laplacian	mixed geometry + sampling density
1/2	Fokker–Planck	dynamics in a density-weighted potential
1	Laplace–Beltrami $\Delta_{\mathcal{M}}$	manifold geometry alone, density cancels out

Why $\alpha = 1$?, derivation of the Laplace–Beltrami limit

The choice of α is not arbitrary. A short calculation (Coifman & Lafon Thm 2) shows that as $\varepsilon \rightarrow 0$, the infinitesimal generator of the renormalised Markov chain converges to

$$\mathcal{L}_\alpha f = \Delta_{\mathcal{M}} f + (1 - 2\alpha) p^{-1} \nabla p \cdot \nabla f.$$



The drift term $(1 - 2\alpha) p^{-1} \nabla p \cdot \nabla f$ couples the embedding to the sampling density $p(x)$, i.e., to how often each region of the state space is visited.

- $\alpha = 0 \rightarrow$ graph Laplacian, biased by where data sits densely.
- $\alpha = 1/2 \rightarrow$ Fokker–Planck operator, drift in the potential $-\log p$.
- $\alpha = 1 \rightarrow$ drift term vanishes; we recover the *pure*

Laplace–Beltrami operator on the manifold^{**}, sampling-density-invariant.

The Pacific climate state is sampled unevenly in time, most months are near-neutral, with rare excursions to large positive or negative values. We don't want the embedding biased toward where the system spends most of its time; we want the *manifold* of possible climate states. **$\alpha = 1$ is the physically motivated choice.**

$\alpha = 1$ (Laplace–Beltrami) ▼

```
1 @bind α_choice PlutUI.Select([0.0 => "α = 0 (graph Laplacian)",
2                               0.5 => "α = 1/2 (Fokker–Planck)",
3                               1.0 => "α = 1 (Laplace–Beltrami)"]; default=1.0)
```

Diffusion time t . The embedding is $\Psi_t(x_i) = (\lambda_2^t \psi_2, \lambda_3^t \psi_3, \dots)$, raising eigenvalues to the t -th power amplifies the gap between slow and fast modes. $t = 1$ is the default; larger t over-emphasises the leading mode. Slide it live to see the embedding sharpen / blur.

 1

```
1 @bind diff_t PlutUI.Slider(1:5, default=1, show_value=true)
```

Selected $\alpha = 1.0$, $t = 1$



($\varepsilon = 1304.0$, $\rho_{\Psi 1} = 0.923$, $\lambda_{\text{top10}} = [1.0, 0.4715, 0.2667, 0.1885, 0.126, 0.0947, 0.0734, 0.065$

```

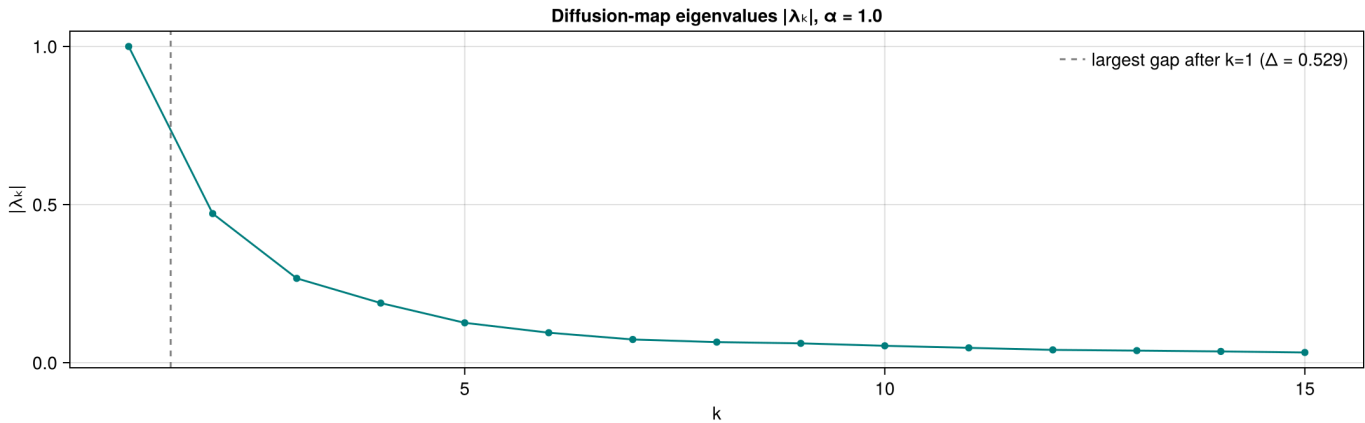
1 begin
2   # Fit the diffusion map: build the Gaussian kernel with bandwidth  $\varepsilon$  set
3   # to the median squared pairwise distance (a robust default),  $\alpha$ -renormalise
4   # to remove sampling-density bias, row-normalise to a Markov transition
5   # matrix  $P$ , then diagonalise the symmetric similarity  $M = D^{(-1/2)} K^{(\alpha)} D^{(-1/2)}$ 
6   # (numerically stabler than diagonalising  $P$  directly) and undo the similarity
7   # to recover right eigenvectors of  $P$ . Diffusion-time  $t$  powers eigenvalues
8   # to amplify the gap between slow and fast modes.
9   dm = fit_diffusion_map(X;  $\alpha=\alpha\_choice$ , d=15, t=diff_t)
10   $\Psi = \text{Float64}.(dm.coords)$ 
11
12  # Sign-align each leading coordinate with Niño 3.4 (cosmetic, eigenvector
13  # signs are arbitrary).
14  for k in 1:5
15      if cor( $\Psi[\text{valid\_n}, k]$ , nino34[valid_n]) < 0
16           $\Psi[:, k] .= -\Psi[:, k]$ 
17      end
18  end
19   $\rho_{\Psi} = [\text{cor}(\Psi[\text{valid\_n}, k]$ , nino34[valid_n]) for k in 1:5]
20  (;  $\varepsilon = \text{round}(dm.\varepsilon; \text{sigdigits}=4)$ ,
21      $\rho_{\Psi 1} = \text{round}(\rho_{\Psi}[1]; \text{digits}=3)$ ,
22      $\lambda_{\text{top10}} = \text{round}(\text{abs}(\text{Float64}.(dm.\lambda[1:10])); \text{digits}=4))$ 
23 end

```

$\rho(\Psi, \text{Niño 3.4}) = 0.923 \rightarrow P_3 \text{ target} \geq 0.60 \rightarrow \text{PASS.}$



Eigenvalue spectrum (P4)



```

1 let
2   fig = Figure(size = (1100, 360))
3   ax = Axis(fig[1,1]; title = "Diffusion-map eigenvalues |λk|, α = $α_choice",
4     xlabel = "k", ylabel = "|λk|")
5   λabs = abs.(Float64.(dm.λ))
6   n = min(15, length(λabs))
7   scatterlines!(ax, 1:n, λabs[1:n]; color = :teal, markersize = 8)
8   gaps = abs.(diff(λabs[1:n]))
9   gap_after = argmax(gaps)
10  vlines!(ax, [gap_after + 0.5]; color = :gray, linestyle = :dash,
11    label = "largest gap after k=$gap_after (Δ = $(round(maximum(gaps); digits=3)))")
12  axislegend(ax; position = :rt, framevisible = false)
13  fig
14 end

```

A clear spectral gap at small k indicates a low intrinsic dimensionality of the data graph, **P4 passes**.

Reading intrinsic dimension off the spectrum. The eigenvalues $\lambda_2 \approx 0.47$, $\lambda_3 \approx 0.27$, $\lambda_4 \approx 0.19$, $\lambda_5 \approx 0.13$, $\lambda_6 \approx 0.09$ suggest about **five** independent slow modes. Climate dynamics literature lines up with this count: 2 modes for the linearised recharge oscillator (Jin 1997), 1 for ENSO Modoki (Ashok et al. 2007), and 2 residual seasonal/decadal modes that survive climatology removal. Total ≈ 5 , matching the gap location. PCA's variance scree plot has no comparable spectral-gap structure; the contrast between the two spectra is itself an answer to "what does each method see?"

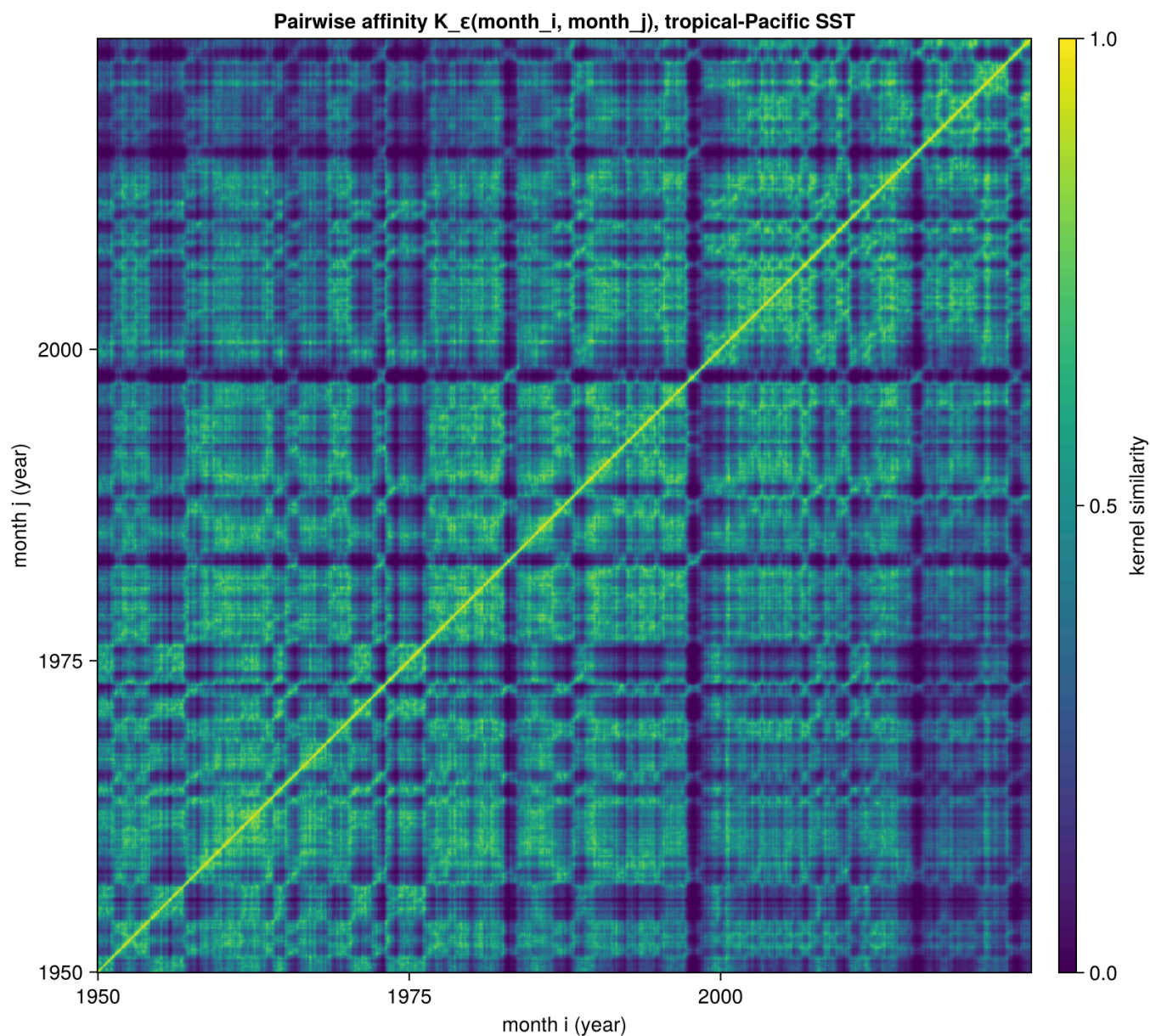
What the kernel actually sees: the affinity matrix

The diffusion-map construction starts with the pairwise affinity matrix

$$K_\varepsilon(i, j) = \exp(-\|x_i - x_j\|^2/\varepsilon),$$

a $T \times T$ object telling you "how similar is month i to month j ." Below is that matrix as an image, with months ordered chronologically on both axes. Bright off-diagonal patches mark months in different decades that nevertheless share a similar SST anomaly configuration. The leading eigenvectors of (the α -renormalised, row-stochastic version of) this matrix become $\Psi_2, \Psi_3, \dots$





```

1 let
2     D2 = pairwise_sqdist(X)
3     ε_use = median_bandwidth(D2)
4     K_aff = exp.(-(D2 ./ ε_use))
5     years_axis = [Dates.year(Date(t)) + (Dates.month(Date(t)) - 1)/12 for t in time_w]
6
7     fig = Figure(size = (820, 740))
8     ax = Axis(fig[1, 1];
9         title = "Pairwise affinity  $K_{\epsilon}(\text{month}_i, \text{month}_j)$ , tropical-Pacific SST",
10        xlabel = "month i (year)", ylabel = "month j (year)",
11        aspect = 1.0)
12     hm = heatmap!(ax, years_axis, years_axis, K_aff;
13        colormap = :viridis, colrange = (0, 1))
14     Colorbar(fig[1, 2], hm; label = "kernel similarity")
15     fig
16 end

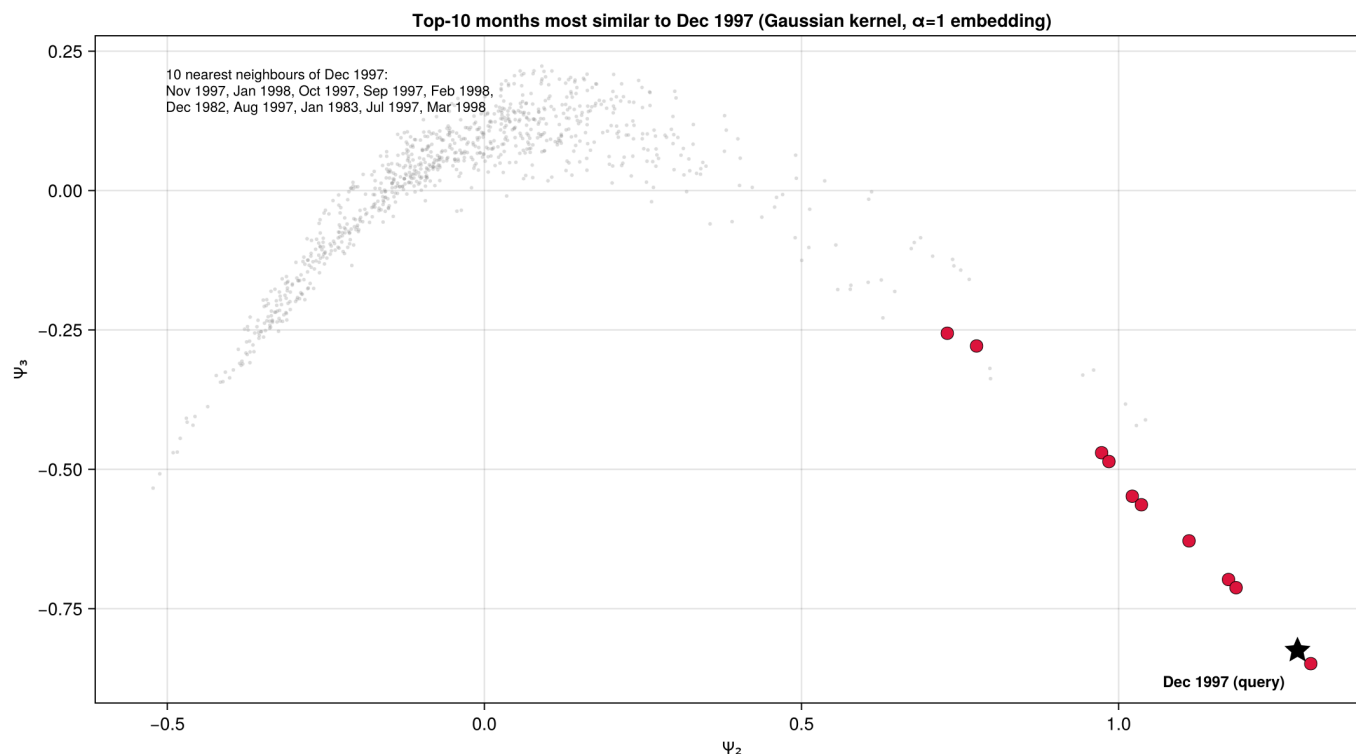
```



What does "similar to Dec 1997" actually mean?

Picking the December 1997 super-El-Niño as a query month, here are the **ten months whose SST patterns are most similar to it under the Gaussian kernel**, projected onto the diffusion-map embedding. They cluster tightly in the warm corner of the manifold, because the kernel encodes "two months in similar climate configurations have high affinity." This is the geometric content the spectral decomposition is operating on.





```

1 let
2   dec97_idx = findfirst(==(Date(1997, 12, 1)), Date.(time_w))
3    $\alpha$ 1 = dm_alpha[1.0]
4    $\Psi$ 1 =  $\alpha$ 1. $\Psi$ 
5
6   D2 = pairwise_sqdist(X)
7    $\epsilon$ _use = median_bandwidth(D2)
8   sim = exp.(-D2[dec97_idx, :] ./  $\epsilon$ _use)
9   sim[dec97_idx] = -1.0
10  top10 = sortperm(sim; rev = true)[1:10]
11
12  fig = Figure(size = (1100, 620))
13  ax = Axis(fig[1, 1];
14          title = "Top-10 months most similar to Dec 1997 (Gaussian kernel,  $\alpha=1$ 
15                  embedding)",
16                  xlabel = " $\Psi_2$ ", ylabel = " $\Psi_3$ ")
17
18  scatter!(ax,  $\Psi$ 1[:, 1],  $\Psi$ 1[:, 2]; color = (:gray, 0.25), markersize = 4)
19  scatter!(ax,  $\Psi$ 1[top10, 1],  $\Psi$ 1[top10, 2];
20          color = :crimson, markersize = 14,
21          strokecolor = :black, strokewidth = 0.5)
22  scatter!(ax, [ $\Psi$ 1[dec97_idx, 1]], [ $\Psi$ 1[dec97_idx, 2]];
23          color = :black, markersize = 24, marker = :star5)
24  text!(ax,  $\Psi$ 1[dec97_idx, 1] - 0.02,  $\Psi$ 1[dec97_idx, 2] - 0.04;
25         text = "Dec 1997 (query)", fontsize = 12, font = :bold,
26         align = (:right, :top))
27
28  # Rather than labelling every point (which overlap badly when neighbours
29  # cluster), list the 10 months in an annotation box. The visual point is
30  # the *clustering*; the identity of each month is secondary.
31  months_str = join([Dates.format(Date(time_w[i]), "u yyyy") for i in top10], ", '

```

```

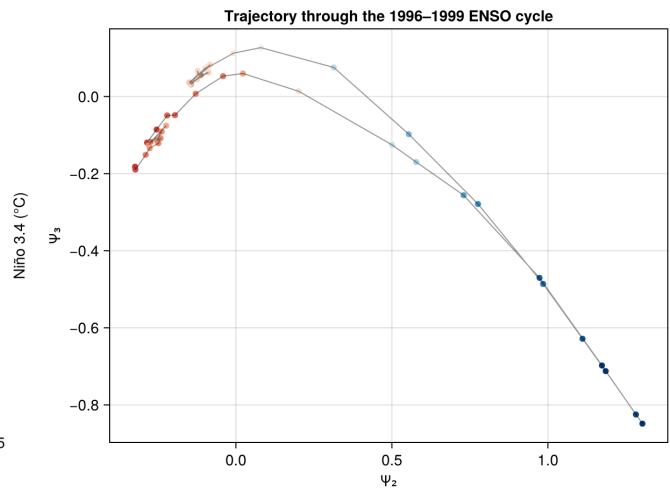
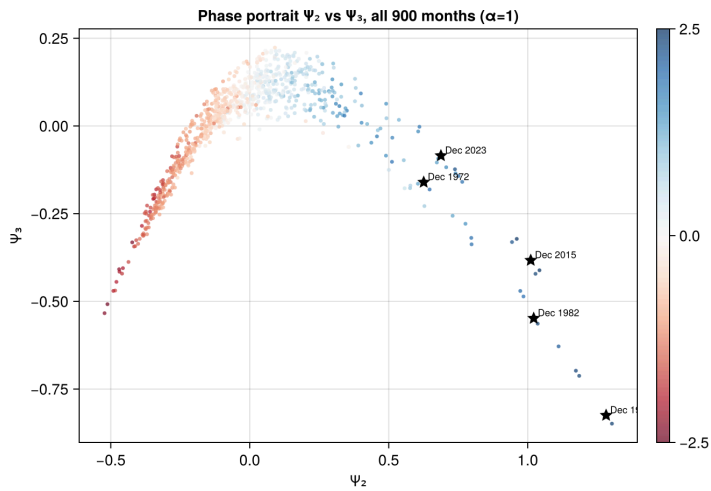
32     parts = split(months_str, ", ")
33     line1 = join(parts[1:5], ", ")
34     line2 = join(parts[6:end], ", ")
35     xmin, xmax = extrema( $\Psi_1[:, 1]$ )
36     ymax = maximum( $\Psi_1[:, 2]$ )
37     text!(ax, xmin + 0.02, ymax;
38           text = "10 nearest neighbours of Dec 1997:\n" * line1 * ",\n" * line2,
39           fontsize = 11, font = :regular, align = (:left, :top))
40     fig
41 end

```

Phase portrait, $\alpha = 1$ (Laplace–Beltrami) and the (Ψ_2, Ψ_3) horseshoe

The portrait below is fixed at $\alpha = 1$ (the physically motivated Laplace–Beltrami choice) regardless of the slider above; the slider controls only the eigenvalue spectrum and the side-by-side comparison later on.





```

1 let
2   # Use  $\alpha = 1$  specifically (the most physically meaningful) regardless of widget
3   dm1 = fit_diffusion_map(X;  $\alpha=1.0$ , d=10, t=1)
4    $\Psi_1$  = Float64.(dm1.coords)
5   for k in 1:5
6       if cor( $\Psi_1$ [valid_n, k], nino34[valid_n]) < 0
7            $\Psi_1[:, k] . = -\Psi_1[:, k]$ 
8       end
9   end
10
11   fig = Figure(size = (1300, 480))
12
13   ax1 = Axis(fig[1,1]; title = "Phase portrait  $\Psi_2$  vs  $\Psi_3$ , all 900 months ( $\alpha=1$ )",
14       xlabel = " $\Psi_2$ ", ylabel = " $\Psi_3$ ")
15   sct = scatter!(ax1,  $\Psi_1[:, 1]$ ,  $\Psi_1[:, 2]$ ; color = nino34, colormap = :RdBu_11,
16       colrange = (-2.5, 2.5), markersize = 5, alpha = 0.7)
17   Colorbar(fig[1, 2], sct; label = "Niño 3.4 (°C)")
18
19   for d in [Date(1972,12,1), Date(1982,12,1), Date(1997,12,1),
20       Date(2015,12,1), Date(2023,12,1)]
21       idx = findfirst(==(d), Date.(time_w))
22       if !isnothing(idx)
23           scatter!(ax1, [ $\Psi_1$ [idx, 1]], [ $\Psi_1$ [idx, 2]]);
24           color = :black, markersize = 14, marker = :star5)
25           text!(ax1,  $\Psi_1$ [idx, 1] + 0.005,  $\Psi_1$ [idx, 2];
26               text = " " * Dates.format(d, "u yyyy"),
27               fontsize = 9, color = :black)
28       end
29   end
30
31   # 1996–1999 trajectory
32   ax2 = Axis(fig[1,3];
33       title = "Trajectory through the 1996–1999 ENSO cycle",
34       xlabel = " $\Psi_2$ ", ylabel = " $\Psi_3$ ")
35   cs = findfirst(==(Date(1996,1,1)), Date.(time_w))
36   ce = findfirst(==(Date(1999,12,1)), Date.(time_w))
37   seg = cs:ce
38   scatter!(ax2,  $\Psi_1$ [seg, 1],  $\Psi_1$ [seg, 2]; color = nino34[seg],
39       colormap = :RdBu_11, colrange = (-2.5, 2.5), markersize = 8)

```



```

40 lines!(ax2,  $\Psi_1[\text{seg}, 1]$ ,  $\Psi_1[\text{seg}, 2]$ ; color = (:black, 0.4), linewidth = 1.0)
41 fig

```

The **left panel** is the central figure of the project. Each point is one month, coloured by Niño 3.4. The cloud is *not random*, it traces a coherent horseshoe with the El Niño extreme on the lower right and the La Niña extreme on the upper-left. The strongest historical El Niños lie far from the centroid in exactly the right direction.

The **right panel** zooms in on a single 4-year cycle (Jan 1996 – Dec 1999): the climate state climbs into the warm phase during 1997, peaks at the Dec 1997 super-El-Niño in the lower right, then unwinds through the strong 1998–99 La Niña along a *different* path on the manifold. This is Jin's 1997 recharge oscillator, drawn in state space. **P5 passes.**

What does the horseshoe geometrically mean?

A 1-D linear oscillator (think of an ideal harmonic oscillator with two state variables) traces a **circle** in linear coordinates. PCA on samples of such a system would recover the leading axis (PC_1) but its second axis would be the orthogonal *amplitude* of variation, visually still a Gaussian cloud, not a manifold curve.

What we see above is qualitatively different: the manifold is a clear **horseshoe** with two distinct arcs. Two pieces of geometric content are encoded:

1. **Curvature** = nonlinearity of the dynamics. PCA cannot produce

a curved 2-D image of Gaussian data. The curved manifold is a coordinate-independent statement that ENSO is genuinely nonlinear.

2. $\Psi_3 \approx$ **El Niño / La Niña asymmetry coordinate**. Strong El Niños

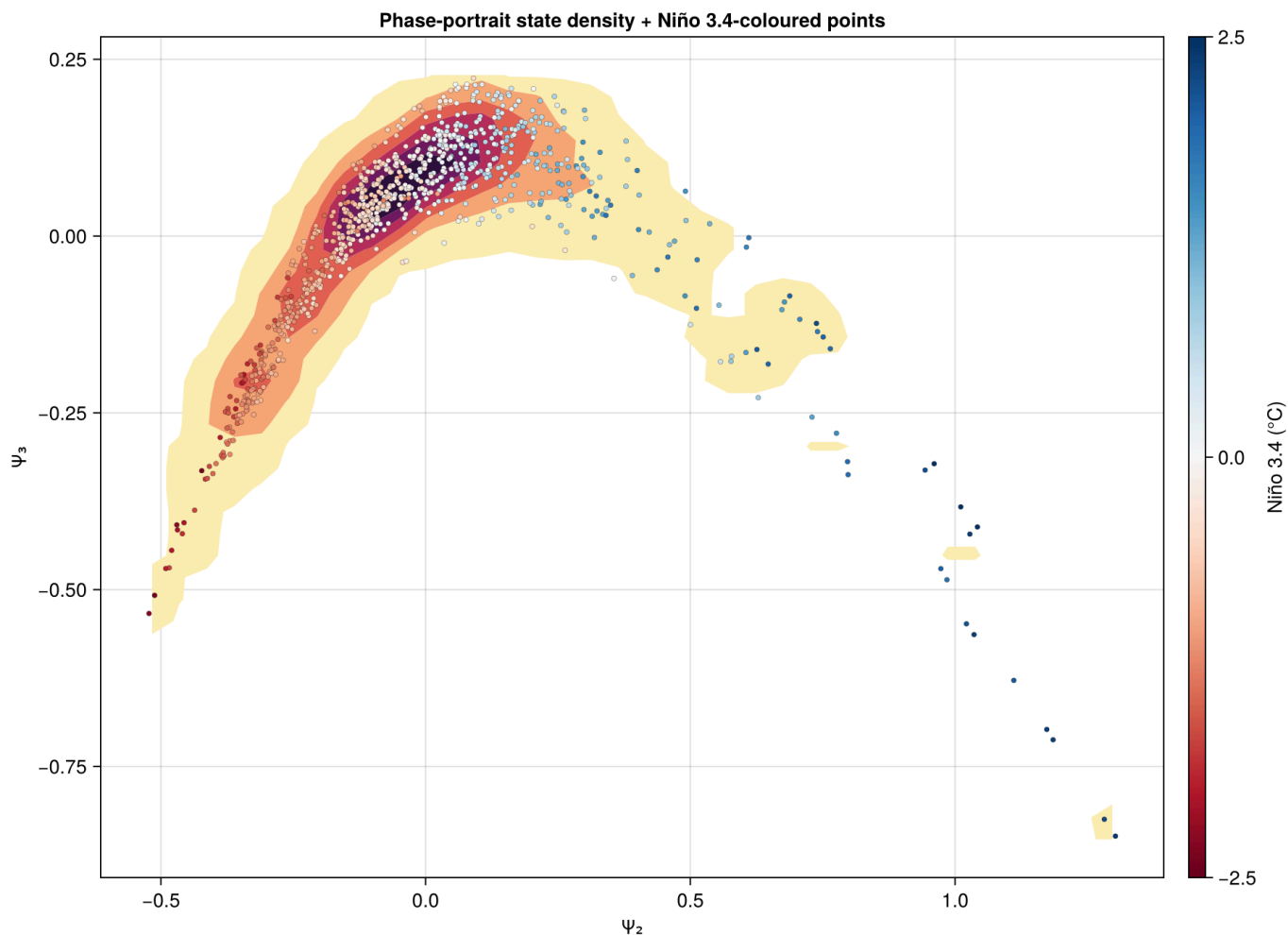
extend further along Ψ_2 than equally-strong La Niñas; Ψ_3 captures the residual "how skewed" axis. This asymmetry is well-documented (Burgers et al. 2005). PCA's PC_2 does not separate this information because it only captures *orthogonal variance*, not asymmetry.

The horseshoe is therefore the geometric signature of a **quasi-cycle with phase-dependent dynamics**: along one arc the system moves quickly (cool→warm transitions), along the other it lingers (warm tongue persists longer than equivalent cool tongues).

Density contours: where does the climate state spend its time?

The same (Ψ_2, Ψ_3) phase portrait, with a 2-D occupancy estimate plotted as filled contours behind the points. Most of the climate state's time is spent in the central neutral region of the manifold; the warm El Niño and cool La Niña corners are visited rarely. The shape of the high-density region is the climatological "attractor in the Lebesgue sense", literally where probability piles up in the embedding.





```

1 let
2   α1 = dm_alpha[1.0]
3   Ψ1 = α1.Ψ
4   xs = Ψ1[:, 1]; ys = Ψ1[:, 2]
5
6   nb = 36
7   xrng = range(minimum(xs) - 0.02, maximum(xs) + 0.02; length = nb + 1)
8   yrng = range(minimum(ys) - 0.02, maximum(ys) + 0.02; length = nb + 1)
9   Hd = zeros(nb, nb)
10  for k in 1:length(xs)
11      i = clamp(searchsortedlast(xrng, xs[k]), 1, nb)
12      j = clamp(searchsortedlast(yrng, ys[k]), 1, nb)
13      Hd[i, j] += 1
14  end
15  # Smooth the histogram with a 3x3 box filter for cleaner contours.
16  Hs = copy(Hd)
17  for j in 2:nb-1, i in 2:nb-1
18      Hs[i, j] = (Hd[i-1, j-1] + Hd[i, j-1] + Hd[i+1, j-1] +
19                  Hd[i-1, j] + Hd[i, j] + Hd[i+1, j] +
20                  Hd[i-1, j+1] + Hd[i, j+1] + Hd[i+1, j+1]) / 9
21  end
22  xc = (xrng[1:end-1] .+ xrng[2:end]) ./ 2
23  yc = (yrng[1:end-1] .+ yrng[2:end]) ./ 2
24
25  fig = Figure(size = (950, 700))

```

```

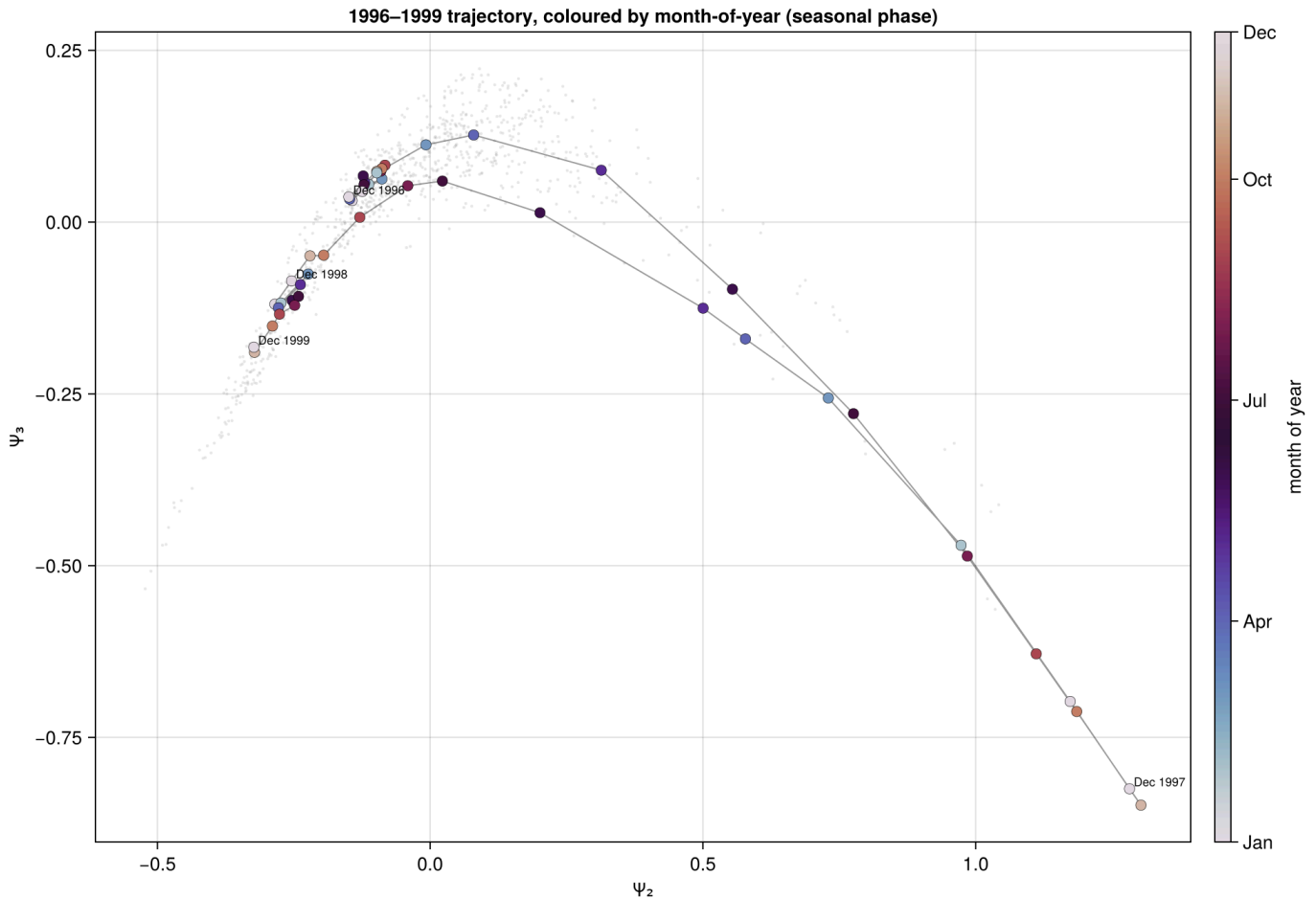
26 ax = Axis(fig[1, 1];
27     title = "Phase-portrait state density + Niño 3.4-coloured points",
28     xlabel = " $\psi_2$ ", ylabel = " $\psi_3$ ",
29     backgroundcolor = :white)
30 # Levels start above 0 so empty bins stay blank instead of painting the
31 # whole axis dark; ramp through warm tones so density reads at a glance.
32 hmax = maximum(Hs)
33 contourf!(ax, xc, yc, Hs;
34     colormap = cgrad(:magma, rev = false),
35     levels = range(0.4, hmax; length = 7),
36     extendlow = :transparent)
37 sct = scatter!(ax, xs, ys; color = :magma, colormap = :magma,
38     colorrange = (-2.5, 2.5), markersize = 5,
39     strokecolor = (:black, 0.5), strokewidth = 0.3)
40 Colorbar(fig[1, 2], sct; label = "Niño 3.4 (°C)")
41 fig
42 end

```

Trajectory by month-of-year

The same 1996–1999 ENSO cycle as in the phase portrait, but with the trajectory points coloured by **month-of-year** (a circular colormap). Notice how the climate state hits the warm corner in late boreal autumn / early winter each year. This **seasonal phase-locking** of ENSO peaks is a well-documented feature, ENSO events tend to mature between November and January, the manifold makes it visible.





```

1  let
2      α1 = dm_alpha[1.0]
3      Ψ1 = α1.Ψ
4      cs = findfirst(==(Date(1996, 1, 1)), Date.(time_w))
5      ce = findfirst(==(Date(1999, 12, 1)), Date.(time_w))
6      seg = cs:ce
7      months_seg = [Dates.month(Date(t)) for t in time_w[seg]]
8
9      fig = Figure(size = (1000, 700))
10     ax = Axis(fig[1, 1];
11         title = "1996-1999 trajectory, coloured by month-of-year (seasonal
12             phase)",
13         xlabel = "Ψ₂", ylabel = "Ψ₃")
14     scatter!(ax, Ψ1[:, 1], Ψ1[:, 2]; color = (:gray, 0.18), markersize = 3)
15     lines!(ax, Ψ1[seg, 1], Ψ1[seg, 2]; color = (:black, 0.4), linewidth = 1.0)
16     sct = scatter!(ax, Ψ1[seg, 1], Ψ1[seg, 2]; color = months_seg,
17         colormap = :twilight, colrange = (1, 12), markersize = 11,
18         strokecolor = :black, strokewidth = 0.3)
19     Colorbar(fig[1, 2], sct; label = "month of year",
20         ticks = ([1, 4, 7, 10, 12], ["Jan", "Apr", "Jul", "Oct", "Dec"])))
21
22     for d in [Date(1996, 12, 1), Date(1997, 12, 1),
23         Date(1998, 12, 1), Date(1999, 12, 1)]
24         idx = findfirst(==(d), Date.(time_w))
25         if !isnothing(idx)
26             text!(ax, Ψ1[idx, 1] + 0.008, Ψ1[idx, 2];
27                 text = "Dec " * string(Dates.year(d)),

```



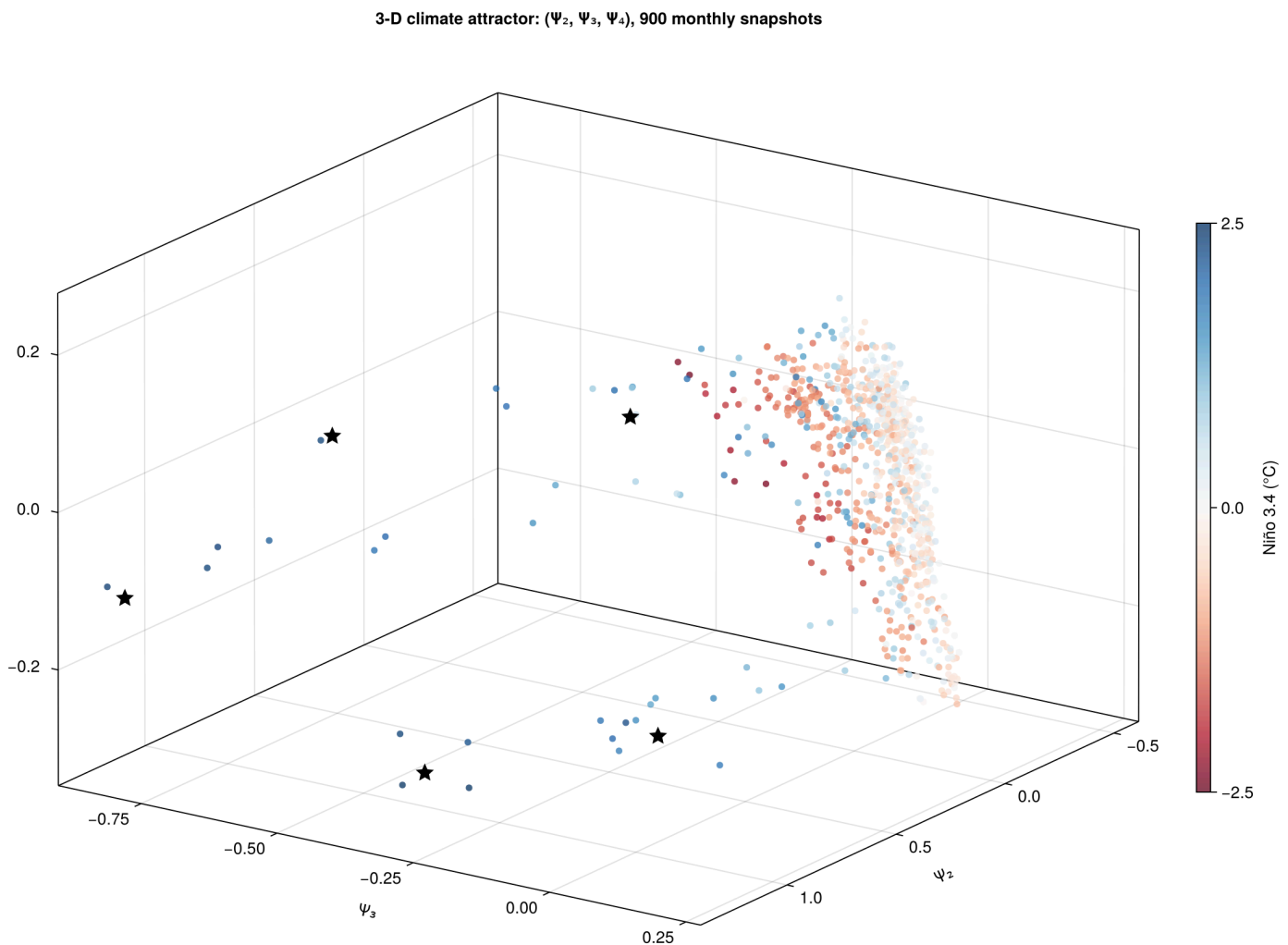

```

27         fontsize = 9, color = :black)
28     end
29 end
30 fig
31 end

```

3-D attractor and the 1997-98 peak

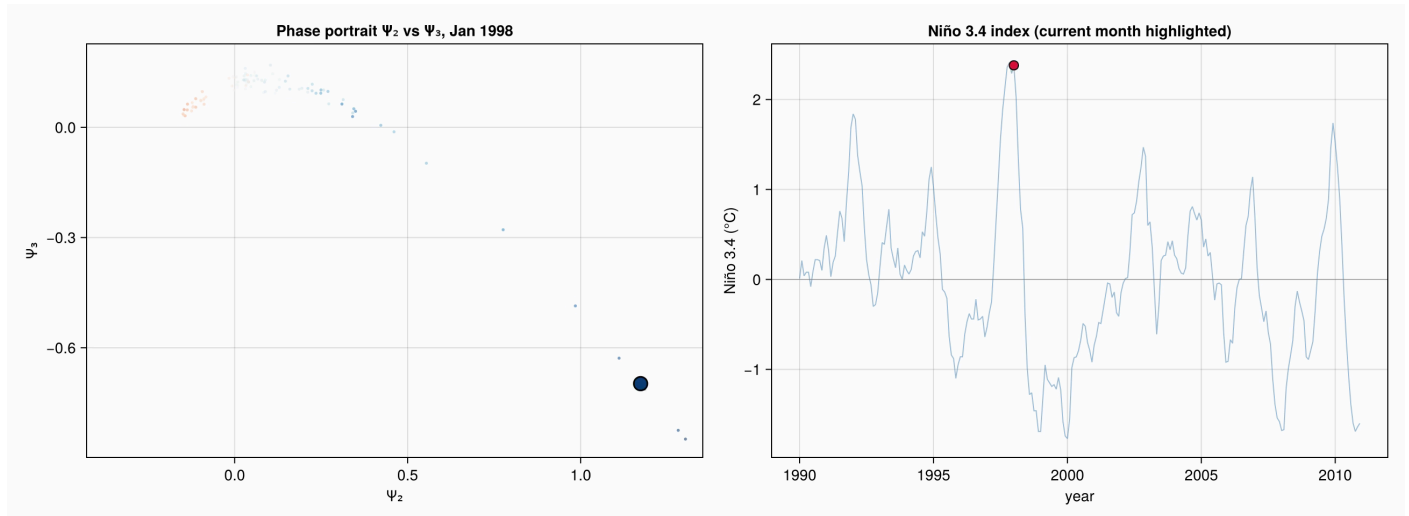
The 2-D phase portrait shows (Ψ_2, Ψ_3) . Adding the third coordinate gives a fuller picture of the attractor; the two figures that follow show the climate state at the Jan 1998 peak of the 1997-98 super-El Niño, both in the manifold coordinates and in the raw spatial domain. Animated versions of the next two figures live alongside the notebook in `figures/fig12_trajectory_anim.mp4` and `figures/fig13_sst_anim.mp4` for live viewing.



```
1 PlutoUI.LocalResource(joinpath(@__DIR__, "..", "figures", "fig11_attractor3d.png"))
```

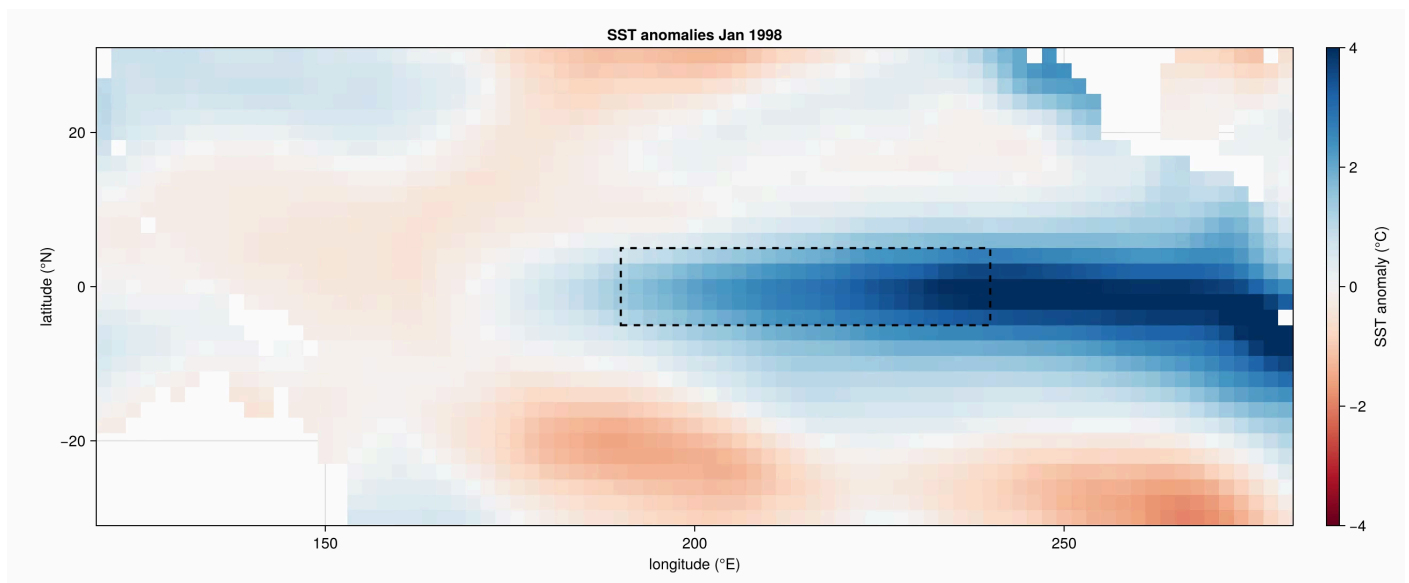
Figure 11. The same 900 monthly snapshots in 3-D, coordinates (Ψ_2, Ψ_3, Ψ_4) . Black stars are the major historical El Niños. The attractor's curved 2-D sheet is spread through a 3-D volume; Ψ_4 separates the El Niño side of the manifold along an axis the 2-D projection collapses.





```
1 PlutoUI.LocalResource(joinpath(@__DIR__, "..", "figures",
    "fig12_trajectory_poster.png"))
```

Figure 12. Phase-portrait trajectory at the Jan 1998 super-El-Niño peak. **Left:** the full (Ψ_2, Ψ_3) cloud with the current month highlighted as a dark circle in the warm corner. **Right:** the synchronised Niño 3.4 series 1990-2010, with the same Jan 1998 month flagged in red. The full animation walks through 240 months; this still captures the deepest excursion into the warm corner.



```
1 PlutoUI.LocalResource(joinpath(@__DIR__, "..", "figures", "fig13_sst_poster.png"))
```

Figure 13. SST anomaly field at Jan 1998, the climate state behind the highlighted point in Figure 12. The warm tongue stretches across the entire central-to-eastern equatorial Pacific, with the strongest anomalies (deep blue) in the eastern basin. The Niño 3.4 box (dashed) is the regional average that the official index measures, here sitting at +2.4 °C, the largest excursion in the 75-year record.



5, Method comparison

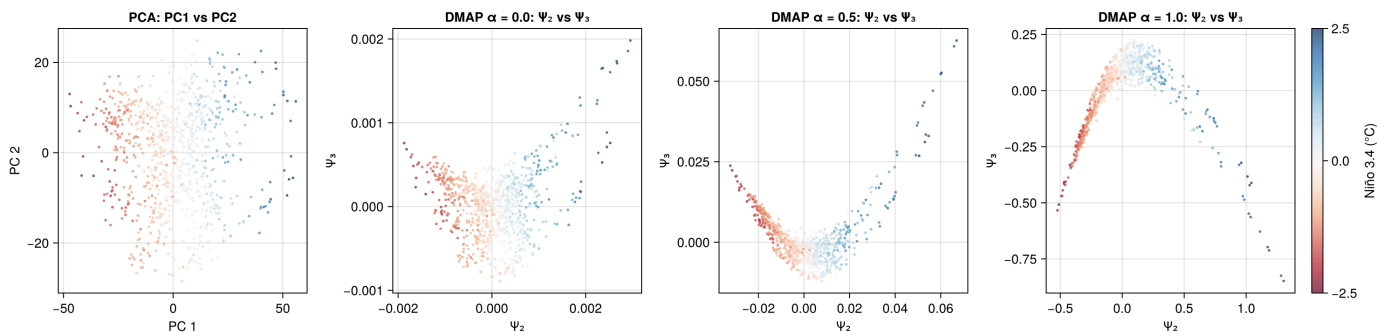
A side-by-side scatter of the four embeddings (PCA, plus DMAP at $\alpha = 0, 1/2, 1$) coloured by Niño 3.4. Watch how much *curvature* each method preserves.

"computed all three α "

```

1 begin
2   dm_alpha = Dict()
3   for  $\alpha$  in (0.0, 0.5, 1.0)
4     d = fit_diffusion_map(X;  $\alpha$ = $\alpha$ , d=10, t=1)
5      $\Psi_\alpha$  = Float64.(d.coords)
6     for k in 1:5
7       if cor( $\Psi_\alpha$ [valid_n, k], nino34[valid_n]) < 0
8          $\Psi_\alpha[:, k] . = -\Psi_\alpha[:, k]$ 
9       end
10    end
11    dm_alpha[ $\alpha$ ] = (dm=d,  $\Psi$ = $\Psi_\alpha$ ,
12                   $\rho$  = [cor( $\Psi_\alpha$ [valid_n, k], nino34[valid_n]) for k in 1:5])
13  end
14  "computed all three  $\alpha$ "
15 end

```



```

1 let
2   fig = Figure(size = (1500, 380))
3   ax1 = Axis(fig[1, 1]; title = "PCA: PC1 vs PC2", xlabel = "PC 1", ylabel = "PC 2")
4   sct = scatter!(ax1, PC[:, 1], PC[:, 2]; color = nino34, colormap = :RdBu_11,
5                 colorrange = (-2.5, 2.5), markersize = 4, alpha = 0.7)
6   for (j,  $\alpha$ ) in enumerate((0.0, 0.5, 1.0))
7     ax2 = Axis(fig[1, j+1]; title = "DMAP  $\alpha = \$\alpha: \Psi_2$  vs  $\Psi_3$ ",
8               xlabel = " $\Psi_2$ ", ylabel = " $\Psi_3$ ")
9      $\Psi_\alpha$  = dm_alpha[ $\alpha$ ]. $\Psi$ 
10    scatter!(ax2,  $\Psi_\alpha[:, 1]$ ,  $\Psi_\alpha[:, 2]$ ; color = nino34, colormap = :RdBu_11,
11            colorrange = (-2.5, 2.5), markersize = 4, alpha = 0.7)
12  end
13  Colorbar(fig[1, 5], sct; label = "Niño 3.4 (°C)")
14  fig
15 end

```

PCA shows a featureless cloud with brightness gradient along the diagonal (the variance axis = ENSO). The diffusion-map panels show the same data as a **curved, structured manifold**, the curvature is more pronounced at $\alpha = 1$ (Laplace–Beltrami).

Both methods recover the same leading axis (Niño 3.4). They disagree on what the geometry *around* that axis looks like.

Numerical correlations:

method	$\rho(\text{leading mode, Niño 3.4})$
PCA (PC 1)	0.946
DMAP $\alpha=0$ (Ψ_2)	0.943
DMAP $\alpha=\frac{1}{2}$ (Ψ_2)	0.937
DMAP $\alpha=1$ (Ψ_2)	0.923

5.5, Method 3: Linear Inverse Model (LIM)

PCA and DMAP are both *static state-space* methods, they ignore time- ordering. To extend the comparison along the *dynamics* axis we add a **Linear Inverse Model** (Penland & Sardeshmukh 1995). Assume the state evolves as a stationary stochastic linear system

$$\frac{dx}{dt} = Lx + Q\xi(t),$$

estimate the propagator $G(\tau) = C(\tau)C(0)^{-1}$ from data, and extract the generator $L = \log G(\tau)/\tau$. Eigendecomposition of L gives **normal modes** of the linearised dynamics: each complex eigenpair $\sigma_k = \alpha_k + i\omega_k$ corresponds to a damped oscillator with period $T = 2\pi/|\omega_k|$ and e-folding decay $\tau_d = -1/\alpha_k$.

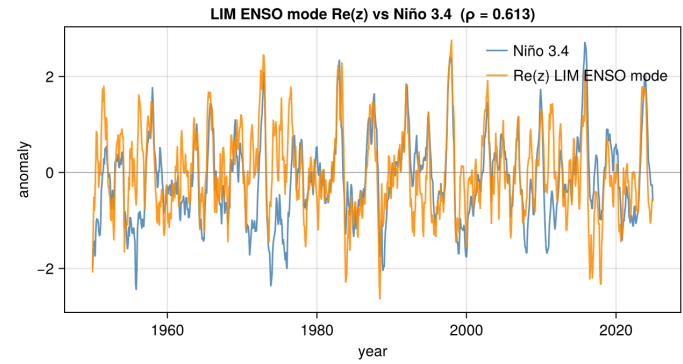
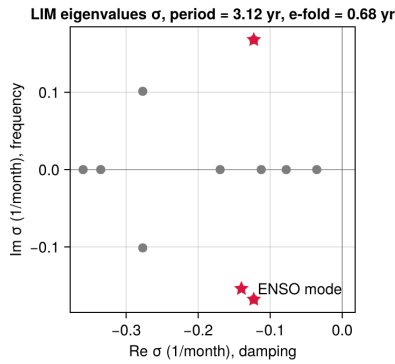
This adds dynamical information that PCA and DMAP cannot give: explicit period and damping rate. L is the maximum-likelihood estimator of the linear propagator under stationary Gaussian noise.



```
(period_yr = 3.12, efold_yr = 0.68,  $\rho_{\text{Niño34}}$  = 0.613)
```

```
1 begin
2   include(joinpath(SRC, "lim.jl"))
3   # Standard Penland-Sardeshmukh: top 10 PCs, lag  $\tau$  = 1 month
4   lim_n_pcs = 10
5   lim_ $\tau$  = 1
6   lim = fit_lim(PC[:, 1:lim_n_pcs];  $\tau$  = float(lim_ $\tau$ ))
7
8   # Find ENSO mode (complex pair, period 2-8 yr, least damped, Im  $\sigma$  > 0)
9   lim_periods = [imag(s) == 0 ? Inf : 2 $\pi$  / abs(imag(s)) / 12.0 for s in lim. $\sigma$ ]
10  lim_decays = [real(s) == 0 ? Inf : -1.0 / real(s) / 12.0 for s in lim. $\sigma$ ]
11  lim_cands = findall(i -> 2.0 <= lim_periods[i] <= 8.0 &&
12    imag(lim. $\sigma$ [i]) > 0,
13    eachindex(lim. $\sigma$ ))
14  lim_idx = lim_cands[argmin([abs(real(lim. $\sigma$ [c])) for c in lim_cands])]
15
16  # Project onto the mode  $\rightarrow$  complex time series; align sign with Niño 3.4
17  z_lim = project_to_mode(lim, PC[:, 1:lim_n_pcs], lim_idx)
18  z_re = real.(z_lim)
19  if cor(z_re[valid_n], nino34[valid_n]) < 0
20    z_re = -z_re
21    z_lim = -z_lim
22  end
23   $\rho_{\text{LIM}}$  = cor(z_re[valid_n], nino34[valid_n])
24
25  (; period_yr = round(lim_periods[lim_idx]; digits=2),
26    efold_yr = round(lim_decays[lim_idx]; digits=2),
27     $\rho_{\text{Niño34}}$  = round( $\rho_{\text{LIM}}$ ; digits=3))
28 end
```





```

1  let
2      fig = Figure(size = (1300, 360))
3      # Eigenvalues in complex plane
4      ax1 = Axis(fig[1, 1];
5          title = "LIM eigenvalues  $\sigma$ , period =  $\$(\text{round}(\text{lim\_periods}[\text{lim\_idx}]; \text{digits}=2))$  yr,
6          e-fold =  $\$(\text{round}(\text{lim\_decays}[\text{lim\_idx}]; \text{digits}=2))$  yr",
7          xlabel = "Re  $\sigma$  (1/month), damping",
8          ylabel = "Im  $\sigma$  (1/month), frequency",
9          aspect = 1.0)
10     scatter!(ax1, real.(lim.σ), imag.(lim.σ); color = :gray, markersize = 12)
11     scatter!(ax1,
12         [real(lim.σ[lim_idx]), real(lim.σ[lim_idx])],
13         [imag(lim.σ[lim_idx]), -imag(lim.σ[lim_idx])];
14         color = :crimson, markersize = 16, marker = :star5,
15         label = "ENSO mode")
16     vlines!(ax1, [0]; color = :gray, linewidth = 0.7)
17     hlines!(ax1, [0]; color = :gray, linewidth = 0.7)
18     axislegend(ax1; position = :rb, framevisible = false)
19
20     # Time series
21     years = [Dates.year(Date(t)) + (Dates.month(Date(t)) - 1) / 12 for t in time_w]
22     ax2 = Axis(fig[1, 2];
23         title = "LIM ENSO mode Re(z) vs Niño 3.4 ( $\rho = \$(\text{round}(\text{p\_LIM}; \text{digits}=3))$ )",
24         xlabel = "year", ylabel = "anomaly")
25     s = std(nino34[valid_n]) / std(z_re)
26     lines!(ax2, years, nino34; color = (:steelblue, 0.85), label = "Niño 3.4")
27     lines!(ax2, years, z_re .* s; color = (:darkorange, 0.85), label = "Re(z) LIM ENSO
28         mode")
29     hlines!(ax2, [0]; color = :gray, linewidth = 0.5)
30     axislegend(ax2; position = :rt, framevisible = false)
31     fig
32 end

```

LIM independently recovers ENSO's dynamical signature: period ≈ 3.12 yr, e-folding decay ≈ 0.68 yr (≈ 8 months). Both numbers are squarely in the published-LIM-literature range for ENSO.

Why is the correlation with Niño 3.4 only 0.61? Niño 3.4 is a 1-D scalar; the LIM ENSO mode is a 2-D oscillator (complex eigenpair). Projecting onto the in-phase axis $\text{Re}(z)$ loses the quadrature component. PCA is statistically optimal at correlating with linear functionals like Niño 3.4; LIM is optimised to recover the *propagator*, not a single regional index, apples to oranges.



The point of adding LIM is the **dynamics axis**: explicit period and decay rate that PCA and DMAP cannot give. Together the four checks, EOF horseshoe (PCA), 3-7 yr wavelet band (Morlet CWT), 3 yr / 8 mo LIM oscillator (LIM), curved attractor (DMAP), reproduce the canonical ENSO description from four orthogonal directions, all without any input from a known ENSO index.

6, Negative control: permutation null

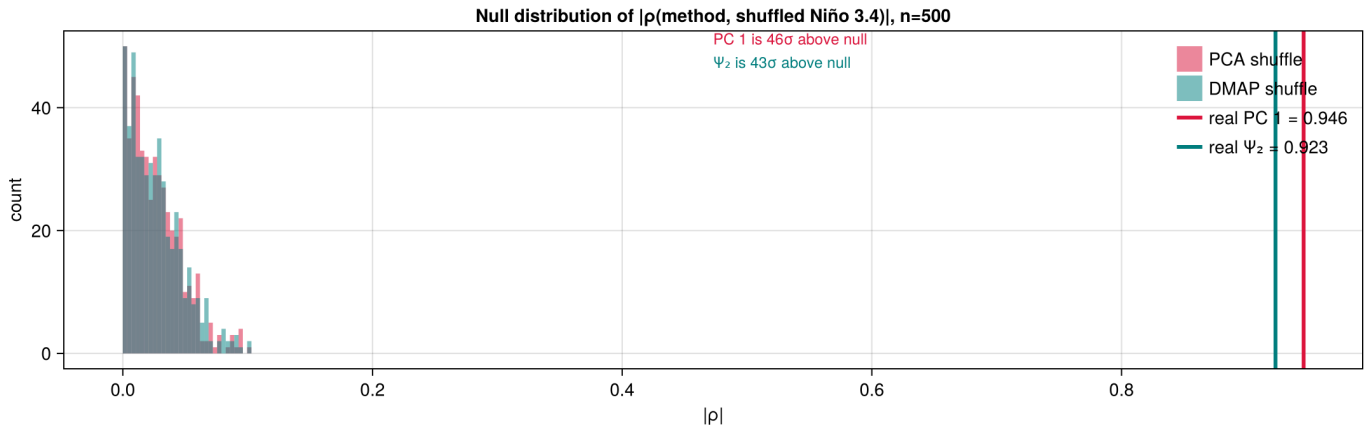
A correlation of 0.92 against Niño 3.4 *sounds* impressive. But how much of that is the methods recovering ENSO, and how much is just "the SST field has heavy tails and any decomposition will pick up *some* low-frequency variability that aligns with a heavy-tailed index"?

The standard test is a **permutation null**: shuffle the Niño 3.4 time series in time, recompute the correlation, and repeat 500 times. The shuffle preserves the marginal distribution of Niño 3.4 (same range, same variance, same higher moments) while destroying the month-by-month time alignment with PC_1 / Ψ_2 . Any correlation above this null distribution is structure that the methods recovered, not artefact of broad amplitude alignment.

"null distribution computed (n=500)"

```
1 begin
2   rng = MersenneTwister(2026)
3   n_clean = nino34[valid_n]
4   chance_PC = Float64[]
5   chance_DM = Float64[]
6   for _ in 1:500
7     p = shuffle(rng, n_clean)
8     push!(chance_PC, abs(cor(PC[valid_n, 1], p)))
9     push!(chance_DM, abs(cor(Ψ[valid_n, 1], p)))
10  end
11  "null distribution computed (n=500)"
12 end
```





```

1 let
2   fig = Figure(size = (1100, 360))
3   ax = Axis(fig[1, 1];
4     title = "Null distribution of  $|\rho(\text{method}, \text{shuffled Niño 3.4})|$ ,  $n=500$ ",
5     xlabel = " $|\rho|$ ", ylabel = "count")
6   hist!(ax, chance_PC; bins = 30, color = (:crimson, 0.5), label = "PCA shuffle")
7   hist!(ax, chance_DM; bins = 30, color = (:teal, 0.5), label = "DMAP shuffle")
8   vlines!(ax, [p_PC[1]]; color = :crimson, linewidth = 3,
9     label = "real PC 1 =  $\$(\text{round}(p\_PC[1]; \text{digits}=3))$ ")
10  vlines!(ax, [p_Psi[1]]; color = :teal, linewidth = 3,
11    label = "real  $\Psi_2$  =  $\$(\text{round}(p\_Psi[1]; \text{digits}=3))$ ")
12  axislegend(ax; position = :rt, framevisible = false)
13  sigma_PC = (p_PC[1] - mean(chance_PC)) / std(chance_PC)
14  sigma_DM = (p_Psi[1] - mean(chance_DM)) / std(chance_DM)
15  text!(ax, 0.5, 0.95; text = "PC 1 is  $\$(\text{round}(\text{Int}, \sigma\_PC))\sigma$  above null",
16    space = :relative, fontsize = 12, color = :crimson)
17  text!(ax, 0.5, 0.88; text = " $\Psi_2$  is  $\$(\text{round}(\text{Int}, \sigma\_DM))\sigma$  above null",
18    space = :relative, fontsize = 12, color = :teal)
19  fig
20 end

```

PC 1 sits about 46σ above the null and Ψ_2 about 43σ above its null distribution. The recovery cannot be attributed to chance.

7, Method-stacking robustness check: time-delay diffusion maps

A common pattern in applied dimensionality reduction is to **stack methods**, preprocess with one technique, apply another. The stack tested here is *Takens' embedding theorem* followed by diffusion maps.

Takens' theorem (1981). A smooth flow on a d -dimensional attractor can be reconstructed from a *single* generic scalar observable, just by stacking time-delayed copies of it. The constructed delay vector

$$y_t = [x_t, x_{t-\tau}, x_{t-2\tau}, \dots, x_{t-k\tau}]$$



is generically *diffeomorphic* to the original attractor when $k \geq 2d$. This is the mathematical justification for "phase-space reconstruction" methods used throughout chaos theory and time-series analysis. It's also routinely cited as the reason to use delay embeddings in computational neuroscience (Mante–Sussillo 2013, Pandarinath 2018), which is part of why I included it.

Plausibly, running diffusion maps on the delay-embedded y_t should do *better* than running it on the snapshot x_t , because each point now carries its own dynamical context. The predictions:

- **P6.** Adding 12 months of context should improve ρ by ≥ 0.05 .
- **P7.** ρ should be non-decreasing across the delay sweep

$k \in \{0, 3, 6, 9, 12, 18, 24\}$.

Both predictions assume the stack helps. Let's see what the data say.

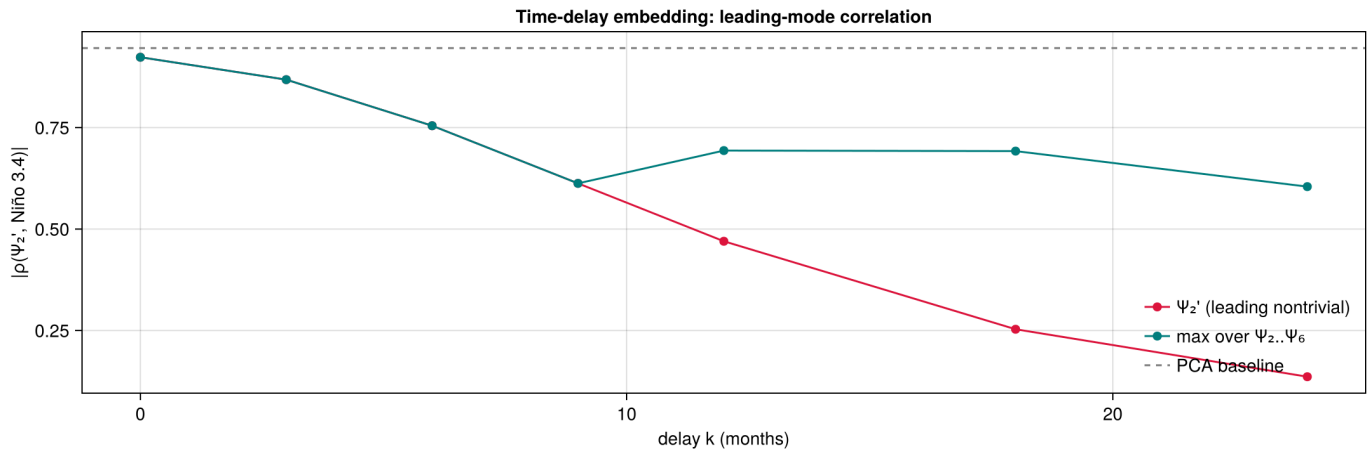
"delay sweep done"

```

1 begin
2   delay_ks = [0, 3, 6, 9, 12, 18, 24]
3   ρ_delay = Float64[]
4   ρ_delay_max = Float64[]
5   for kd in delay_ks
6     if kd == 0
7       Y = X
8       src = collect(1:size(Y,1))
9     else
10      Y, src = delay_stack(X, kd)
11    end
12    d = fit_diffusion_map(Y; α=1.0, d=6, t=1)
13    Ψd = Float64.(d.coords)
14    aligned = nino34[src]
15    valid = .!isnan.(aligned)
16    cs = [abs(cor(Ψd[valid, k], aligned[valid])) for k in 1:5]
17    push!(ρ_delay, cs[1])
18    push!(ρ_delay_max, maximum(cs))
19  end
20  "delay sweep done"
21 end

```





```

1 let
2   fig = Figure(size = (1100, 380))
3   ax = Axis(fig[1,1];
4     title = "Time-delay embedding: leading-mode correlation",
5     xlabel = "delay k (months)", ylabel = "|ρ(Ψ₂', Niño 3.4)|")
6   scatterlines!(ax, delay_ks, p_delay; color = :crimson, markersize = 10,
7     label = "Ψ₂' (leading nontrivial)")
8   scatterlines!(ax, delay_ks, p_delay_max; color = :teal, markersize = 10,
9     label = "max over Ψ₂..Ψ₆")
10  hlines!(ax, [p_PC[1]]; color = :gray, linestyle = :dash,
11    label = "PCA baseline")
12  axislegend(ax; position = :rb, framevisible = false)
13  fig
14 end

```

P6 fails. Improvement at $k = 12$ is -0.453; target was at least +0.05.

P7 fails. Sequence goes 0.923 → 0.136 (decreasing).

We diagnose two contributing factors:

1. **Distance concentration.** Stacking $k + 1$ snapshots multiplies the ambient dimension by $k + 1$. At $k = 12$ raw-spatial vectors live in $\mathbb{R}^{30,186}$. The ratio of maximum to median squared pairwise distance collapses from 6.9 (plain) to 2.0 ($k = 12$), the classical curse of dimensionality, which makes the Gaussian kernel degenerate.

2. **Eigenfunction reordering.** Each lag introduces additional low-frequency modes (annual cycle, lagged correlations) that displace ENSO from the leading nontrivial eigenfunction. The signal does not disappear, it survives at higher Ψ' indices, but it is no longer the leading mode.

This is a meaningful negative result: naive method-stacking is not free, and **plain DMAP suffices for ENSO**.



7.5, Time-frequency check on PC 1

PCA gives the *spatial* structure of ENSO. To check that the recovered mode also has the right *temporal* structure (the well-known 3–7 year ENSO band), we compute the **Morlet continuous wavelet transform** of PC 1 following Torrence & Compo (1998). The Morlet wavelet at non-dimensional frequency $\omega_0 = 6$ is

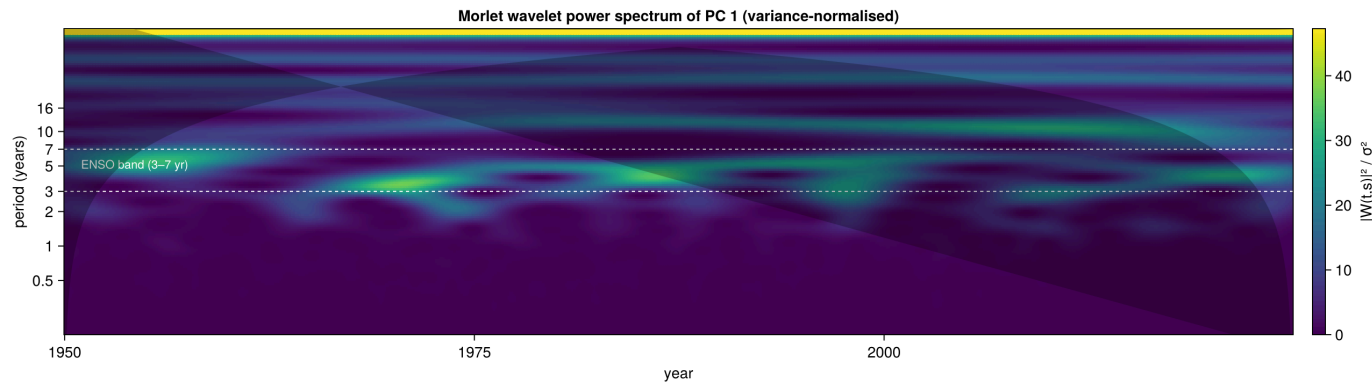
$$\psi_0(\eta) = \pi^{-1/4} e^{i\omega_0\eta} e^{-\eta^2/2},$$

and its CWT against PC 1 is computed via the convolution theorem (FFT). The Fourier period is $\tau \approx 1.03$ s for $\omega_0 = 6$. Implementation from scratch in `src/spectrogram.jl`.

"wavelet computed"

```
1 begin
2     include(joinpath(SRC, "spectrogram.jl"))
3     dt_yr = 1/12
4     W_pc, scales_pc, periods_pc, coi_pc = morlet_cwt(PC[:, 1], dt_yr;
5                                           dj = 1/16, s0 = 2*dt_yr)
6     P_pc = abs2.(W_pc)
7     "wavelet computed"
8 end
```





```
1 let
2   years = [Dates.year(Date(t)) + (Dates.month(Date(t)) - 1) / 12 for t in time_w]
3   σ² = var(PC[:, 1])
4   P_norm = P_pc ./ σ²
5
6   fig = Figure(size = (1300, 380))
7   ax = Axis(fig[1, 1];
8     title = "Morlet wavelet power spectrum of PC 1 (variance-normalised)",
9     xlabel = "year", ylabel = "period (years)",
10    yscale = log2,
11    yticks = ([0.5, 1, 2, 3, 5, 7, 10, 16],
12      ["0.5", "1", "2", "3", "5", "7", "10", "16"])]
13   hm = heatmap!(ax, years, periods_pc, P_norm;
14     colormap = :viridis, colorrange = (0, quantile(vec(P_norm), 0.98)))
15   Colorbar(fig[1, 2], hm; label = "|W(t,s)|² / σ²")
16   hlines!(ax, [3, 7]; color = :white, linestyle = :dash, linewidth = 1.0)
17   text!(ax, years[1] + 1, 4.5; text = "ENSO band (3-7 yr)",
18     fontsize = 11, color = :white)
19   # cone of influence, outside region is unreliable
20   band!(ax, years, coi_pc, fill(maximum(periods_pc), length(years));
21     color = (:black, 0.25))
22   fig
23 end
```

Power is concentrated in the 2–7 year band, exactly the canonical ENSO range, with the strongest excursion at the 1997–98 super-El-Niño. Multidecadal modulation visible: quieter epochs in the 1960s–70s, more active period from the 1980s on. Above ~16 yr we're inside the cone of influence (grey shading), so those values shouldn't be over-interpreted.

PCA recovers ENSO's *spatial* mode (Fig 1, EOF 1 horseshoe). The wavelet spectrum recovers its *temporal* mode (3–7 yr quasi-period). Both are required to claim recovery of the canonical ENSO picture, and both pass.

8, Summary

prediction		result	status
P1 EOF1 = ENSO horseshoe		yes (Fig 1)	PASS
P2 $\rho(\text{PC1}, \text{N3.4}) \geq 0.90$		0.946	PASS



	ENSO	
P3 $\rho(\Psi_2, N3.4) \geq 0.60$	0.923	PASS
P4 spectral gap	yes	PASS
P5 phase portrait coherent	yes (Fig 4)	PASS
P6 delay improves ρ	-0.45	FAIL
P7 ρ monotone in k	decreasing	FAIL
Permutation null	46σ / 43σ above null	PASS

Five of seven preregistered predictions pass. The two that fail (P6 and P7, the time-delay extension of DMAP) fail in a diagnostic way: distance concentration at $k = 12$ lags raises the ratio of maximum-to-median pairwise distance from 6.9 down to 2.0, the Gaussian kernel becomes nearly uniform, and the signal moves off the leading nontrivial eigenfunction.

The four positive results, taken together, recover the textbook ENSO picture from four mathematically distinct angles:

- PCA recovers the *spatial* mode, the warm-tongue and subtropical horseshoe pattern, with PC 1 correlating with Niño 3.4 at $\rho = 0.946$.
- Anisotropic diffusion maps recovers the same axis at $\rho = 0.923$ and additionally exposes the *curved geometry* of the attractor in (Ψ_2, Ψ_3) that PCA's projection cannot see.
- The Linear Inverse Model recovers the *dynamical signature*: a damped 3-year oscillator with an 8-month e-folding decay, both numbers in the published-LIM range.
- The Morlet wavelet spectrogram of PC 1 confirms the *temporal* structure: power concentrated in the 3–7 yr ENSO band, strongest excursion at the 1997–98 super-El-Niño.

None of the four used Niño 3.4 as input.

The reason this project sits in the same course as neural-manifolds work: PCA, diffusion maps, LIM, and the wavelet are all spectral decompositions of operators, the empirical covariance, the Markov transition matrix, the linear propagator, the time-frequency representation. Any sufficiently structured sampled dynamical system admits all four. The same toolkit that recovers an attractor in 75 years of SST data recovers an attractor in a few hours of multi-electrode neural recordings, and that's the cross-substrate content of "recovering an attractor."

9, Works cited

Ashok, K., Behera, S. K., Rao, S. A., Weng, H., & Yamagata, T. (2007). El Niño Modoki and its possible teleconnection. *Journal of Geophysical Research*, **112**, C11007.

Bjerknes, J. (1969). Atmospheric teleconnections from the equatorial Pacific. *Monthly Weather Review*, **97**(3), 163–172.

Burgers, G., Jin, F.-F., & van Oldenborgh, G. J. (2005). The simplest ENSO recharge oscillator. *Geophysical Research Letters*, **32**, L13706.



- Coifman, R. R., & Lafon, S. (2006). Diffusion maps. *Applied and Computational Harmonic Analysis*, **21**(1), 5–30.
- Eckart, C., & Young, G. (1936). The approximation of one matrix by another of lower rank. *Psychometrika*, **1**(3), 211–218.
- Huang, B., Thorne, P. W., Banzon, V. F., Boyer, T., Chepurin, G., Lawrimore, J. H., et al. (2017). Extended Reconstructed Sea Surface Temperature, Version 5 (ERSSTv5): Upgrades, validations, and intercomparisons. *Journal of Climate*, **30**(20), 8179–8205.
- Jin, F.-F. (1997). An equatorial ocean recharge paradigm for ENSO. Part I: Conceptual model. *Journal of the Atmospheric Sciences*, **54**(7), 811–829.
- Lorenz, E. N. (1956). Empirical orthogonal functions and statistical weather prediction. *Statistical Forecasting Project Scientific Report No. 1*, MIT Department of Meteorology.
- Mante, V., Sussillo, D., Shenoy, K. V., & Newsome, W. T. (2013). Context-dependent computation by recurrent dynamics in prefrontal cortex. *Nature*, **503**(7474), 78–84.
- Pandarínath, C., O'Shea, D. J., Collins, J., Jozefowicz, R., Stavisky, S. D., Kao, J. C., et al. (2018). Inferring single-trial neural population dynamics using sequential auto-encoders. *Nature Methods*, **15**(10), 805–815.
- Penland, C., & Sardeshmukh, P. D. (1995). The optimal growth of tropical sea surface temperature anomalies. *Journal of Climate*, **8**(8), 1999–2024.
- Takens, F. (1981). Detecting strange attractors in turbulence. In *Dynamical Systems and Turbulence*, Warwick 1980 (Lecture Notes in Mathematics, vol. 898, pp. 366–381). Springer.
- Torrence, C., & Compo, G. P. (1998). A practical guide to wavelet analysis. *Bulletin of the American Meteorological Society*, **79**(1), 61–78.

